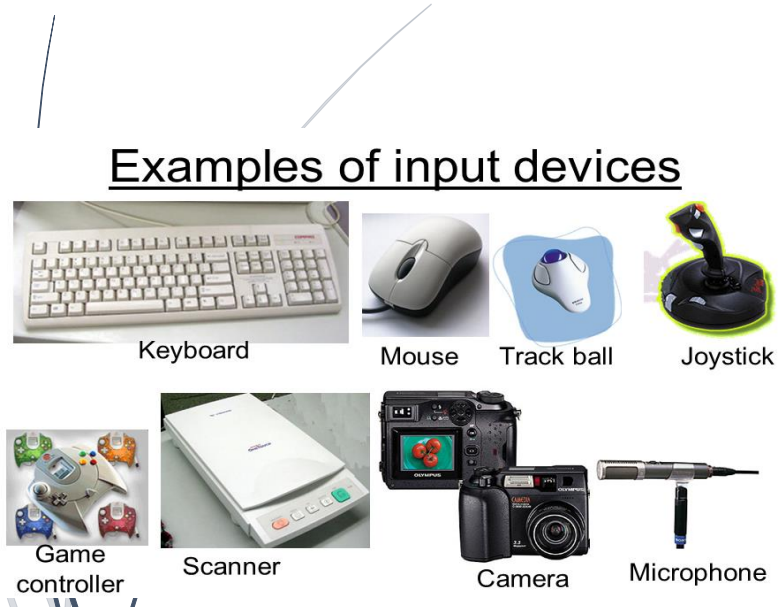
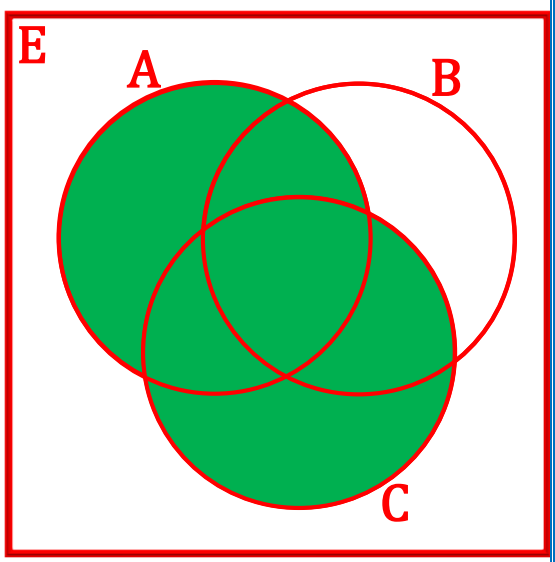
MATHEMATICS 4024
GRADE 10
TEACHING NOTES TOPIC BY TOPIC



A Comprehensive Study and Step-by-Step Solutions to ECZ Mathematics 4024

A Resource for Mastering Grade 10 Mathematics Exam Questions from ECZ Past Papers (2018–2024)





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SETS

Objectives

By the end of this topic, learners should

1. Define what a set is and provide examples.
2. Identify and explain the different types of sets (finite, infinite, empty).
3. Understand the concept of subsets and proper subsets.
4. Perform basic set operations (union, intersection, difference).
5. Apply set theory to real-world problems.

Definition

- A set is an unordered collection of unique objects, known as elements or members.
- Sets can contain numbers, letters, symbols, or even other sets.
- In mathematics, a set is defined as a collection of distinct, well-defined objects forming a group.
- The symbols used when dealing with sets are called set notations.
- The table below shows some common set notations and their meanings.

SYMBOL	NAME	MEANING
$\cup$	Union	Combining all the given sets
$\cap$	Intersection	What is common in both sets
$'$	Complement	What is out of the set or elements that does not found / exist in a given set

REAL NUMBER SYSTEM

- In mathematics, a real number is a number that can be used to measure a continuous one-dimensional quantity such as a distance, duration or temperature.
- However, it is very important to understand all the types of numbers in order to do well in sets specifically in paper 1.
- Natural numbers are common counting numbers starting from 1 up to infinite, e.g.
  {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, ...}
- Whole numbers are the set of natural numbers including with the number 0 adjoined, e.g.
  {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, ...}
- Prime numbers are natural numbers greater than 1 which has only 1 and itself as factors, e.g. {2, 3, 5, 7, 11, 13, 17, 19, 23, ...}
- Composite number is a natural number greater than 1 which has more factors than 1 and itself.
- Integers are whole numbers with their opposites (negative numbers) adjoined.
- Rational numbers are all numbers which can be written as fractions.
- Irrational numbers are all numbers which cannot be written as fractions.
- Real numbers are the set of rational numbers with the set of irrational numbers adjoined.
- Complex number is a number which can be written in the form $a + bi$ where a and b are real numbers and i is the square root of -1.

- An Even number is a number which can be exactly divisible by 2.
- A factor of a number is a number which can be exactly divisible by that number.
- Multiples of a number are numbers formed by multiplying a number by other numbers starting from 1.

Key Properties of Sets

1. Unordered: Sets do not have a specific order or arrangement.
2. Unique Elements: Sets only contain distinct elements, with no duplicates.
3. Finite or Infinite: Sets can have a finite or infinite number of elements.

Types of Sets

1. Empty Set: A set with no elements, denoted as {} or ∅.
2. Singleton Set: A set with only one element, denoted as {a}.
3. Finite Set: A set with a finite number of elements, denoted as {a, b, c, ...}.
4. Infinite Set: A set with an infinite number of elements, denoted as {a, b, c, ... ∞}.
5. Subsets: A set whose elements are all contained within another set.

Set Operations

1. Union: Combines elements from two or more sets.
2. Intersection: Finds common elements between two or more sets.
3. Difference: Removes elements from one set that are present in another set.
4. Complement: Finds elements not present in a given set.

Notations and Symbols

1. Set Notation: {a, b, c, ...}
2. Element Symbol: ∈ (element of)
3. Subset Symbol: ⊆ (subset of)
4. Union Symbol: ∪ (union of)
5. Intersection Symbol: ∩ (intersection of)

Venn diagrams

- A Venn diagram is a visual representation of sets and their relationships using overlapping circles or ellipses.

Components of a Venn Diagram

1. Sets: Represented by circles or ellipses.
2. Elements: Represented by points or labels within the circles.
3. Overlapping regions: Represent the intersection of sets.
4. Non-overlapping regions: Represent the difference between sets.

Types of Venn Diagrams

1. Two-circle Venn diagram: Used to represent two sets and their relationships.
2. Three-circle Venn diagram: Used to represent three sets and their relationships.
3. Multi-circle Venn diagram: Used to represent more than three sets and their relationships.

Uses of Venn Diagrams

1. Set theory: Venn diagrams help visualize set operations, such as union, intersection, and difference.
2. Logic: Venn diagrams are used to represent logical relationships between statements.
3. Computer science: Venn diagrams are used to visualize data relationships, such as database schema and data flow.
4. Statistics: Venn diagrams are used to visualize data distributions and relationships.

How to Read a Venn Diagram

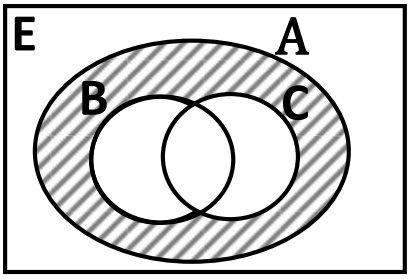
1. Identify the sets: Look for the circles or ellipses representing the sets.
2. Identify the elements: Look for the points or labels within the circles representing the elements.
3. Identify the overlapping regions: Look for the regions where the circles overlap, representing the intersection of sets.
4. Identify the non-overlapping regions: Look for the regions where the circles do not overlap, representing the difference between sets.

Common Venn Diagram Symbols

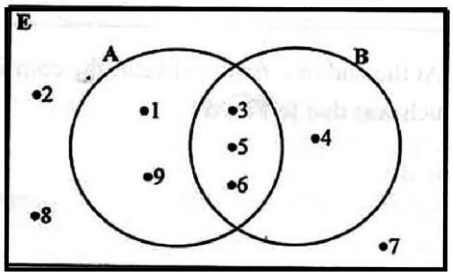
1. ∪ (Union): Represents the combination of two or more sets.
2. ∩ (Intersection): Represents the overlap between two or more sets.
3. ∁ (Complement): Represents the elements not in a set.
4. ⊆ (Subset): Represents a set that is contained within another set.

2018.p1.Oct

(a) Use set notation to describe the shaded region in the diagram below.



(b) The Venn diagram below shows sets A and B. List $A' \cap B$



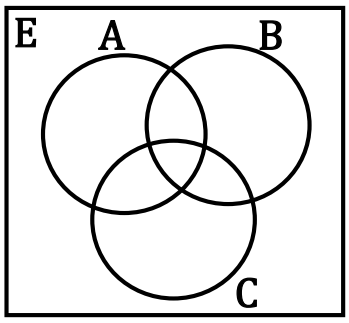
Solution

(a) $(B \cup C)' \cap A$

(b) $A' \cap B = \{4\}$

2019.P1.Nov

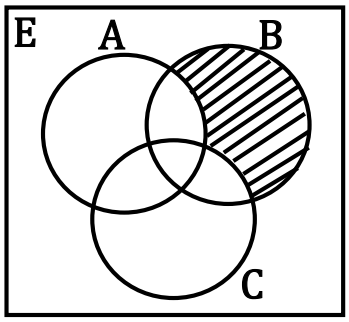
(a) Shade $(A \cup C)' \cap B$ in the Venn diagram below.



(b) Set A = {Prime numbers less than 12}. List set A.

Solution

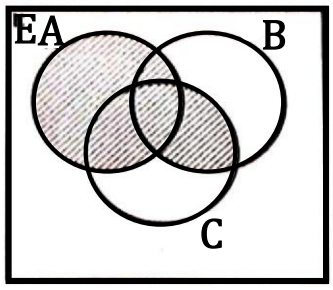
(a)



(b) A = {2, 3, 5, 7, 11}

2021.p1.Nov

(a) Use set notation to describe the shaded part in the Venn diagram below.



(b) It is given that the universal set E = {first 8 whole numbers} and A = {even numbers}. List set A.

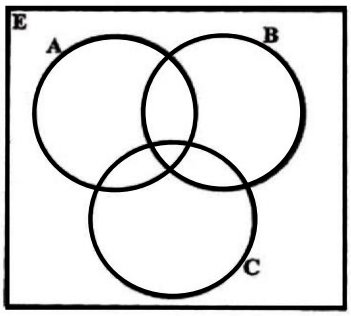
Solution

(a) $(B \cap C) \cup A$

(b) S = {2, 4, 6}

2022.p1.Nov

(a) Shade $(A \cup B') \cap C$ on the diagram below.



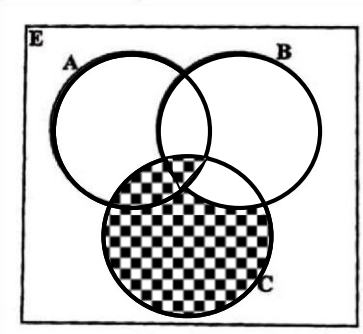
(b) Given that the universal set $E = \{1, 3, 5, 7, 9, 11\}$, $X = \{1, 5, 9\}$ and $Y = \{3, 9, 11\}$, list $X \cap Y'$.

Solution

(a) $Y' = \{1, 5, 7\}$

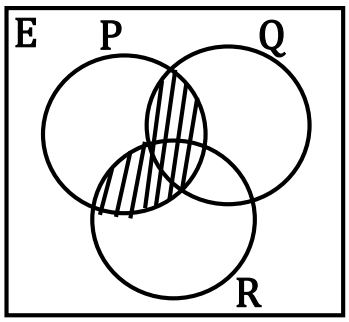
$X \cap Y' = \{1, 5\}$

(b)



2023.P1.Nov

(a) The Venn diagram below shows three sets P, Q and R. Use set notation to describe the shaded part.



(b) Set A = $\{x: 1 < x \leq 15, x \text{ is a prime number}\}$ and set B = $\{x: 0 \leq x < 10, x \text{ is an odd number}\}$. List the set A ∩ B.

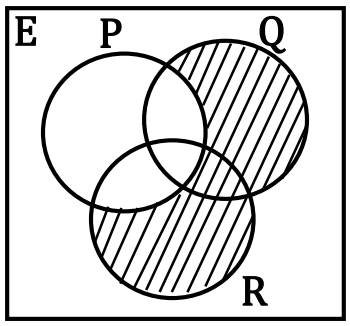
Solution

(a) $(P \cap Q) \cup (P \cap R)$
$P \cap (Q \cup R)$

(b) $A = \{2, 3, 5, 7, 11, 13\}$
$B = \{1, 3, 5, 7, 9\}$
$A \cap B = \{3, 5, 7\}$

2024.p1.GCE

(a) Use set notation to describe the shaded part in the Venn diagram below.



(b) It is given that the universal set $E = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $P = \{2, 4, 6, 8\}$ and $Q = \{2, 3, 5, 7\}$. List $P' \cap Q$.

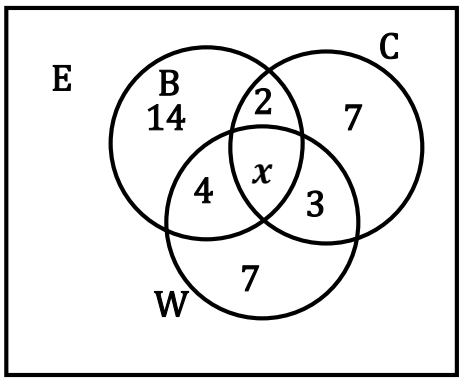
Solution

(a) $(Q \cup R) \cap P'$

(b) $P' \cap Q = \{3, 5, 7\}$

2018.P2.GCE

The diagram below shows how learners in a Grade 12 class at Twaenda School travel to school. The learners use either buses (B), car (C) or walk (W) to school.



(i) If 22 learners walk to school, find the value of x. [2]

(ii) How many learners use

(a) only one mode of transport, [1]

(b) two different modes of transport? [1]

Solution

(i) $x + 3 + 4 + 7 = 22$

$x + 14 = 22$

$x = 22 - 14$

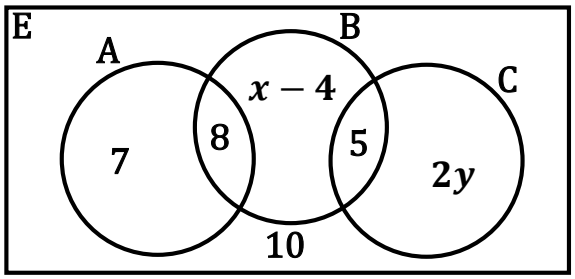
$x = 8$

(ii) (a) $14 + 7 + 7 = 28$ learners

(b) $4 + 2 + 3 = 9$ learnes

2019.P2.Nov

The Venn diagram below shows the number of elements in sets A, B and C.



Find

(i) x, such that $n(B) = n(B \cup C)'$, [2]

(ii) y, such that $n(C) = n(A)$, [1]

(iii) $n(E)$, [1]

(iv) $n(B')$. [1]

Solution

(i) $n(B) = n(B \cup C)'$

$x - 4 + 8 + 5 = 7 + 10$

$x + 9 = 17$

$x = 17 - 9$

$x = 8$

(ii) $n(C) = n(A)$

$2y + 5 = 8 + 7$

$2y + 5 = 15$

$2y = 15 - 5$

$2y = 10$

$y = \frac{10}{2}$

$y = 5$

(iii) $n(E) = 44$

(iv) $n(B)' = 27$

2020.P2.Nov

Of the 115 students who attended an end of year party, 74 took fanta, 93 took sprite, 87 took coke, 61 took fanta and sprite, 71 took sprite and coke, 60 took fanta and coke and 50 took all the three drinks.

(i) Illustrate this information on a Venn diagram. [2]

(ii) How many students took

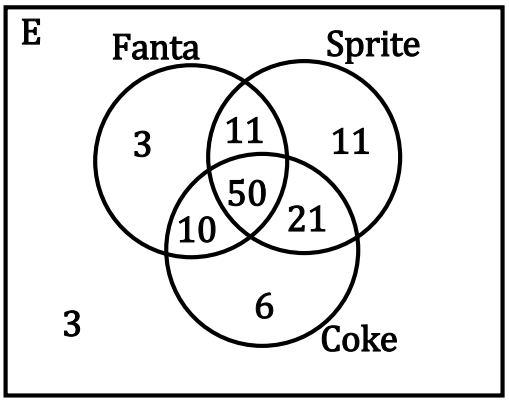
(a) none of the drinks, [1]

(b) fanta and sprite but not coke, [1]

(c) at least two different drinks? [1]

Solution

(i)



(ii) (a) 3 students

(b) 11 students

(c) 92 students

2021.P2.Nov

At a certain Secondary School in Grade 12 all the learners take at least one of the following optional subjects: French (F), Home Economics (HE) and Geography (G). 20 leaners take French only, 25 learners take Home Economics only and 22 learners take Geography only. Furthermore, 9 learners take French and Geography, 12 learners take French and Home Economics, 15 leaners take Geography and Home Economics and 4 learners take all the three subjects.

(i) Illustrate this information on a Venn diagram. [2]

(ii) How many learners

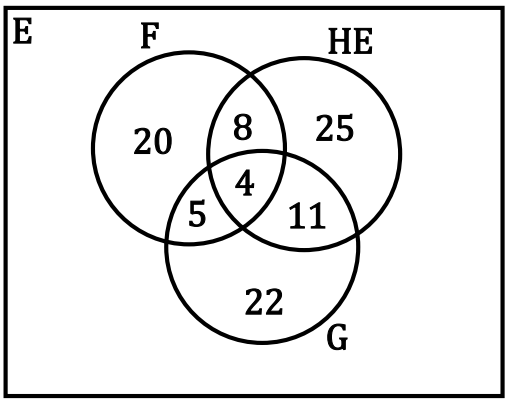
(a) were at this school, [1]

(b) take Home Economics and Geography but not French, [1]

(c) take two optional subjects only? [1]

Solution

(i)



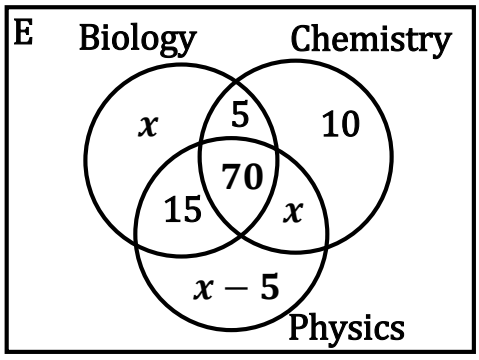
(ii) (a) 95 learners

(b) 11 learners

(c) 24 learnes

2022.P2.Nov

The diagram below shows the number of learners who took at least one of the three subjects Biology, Chemistry and Physics.



(i) Given that the total number of learners was 170, calculate the value of x. [2]

(ii) How many learners took

(a) Biology and Physics but not Chemistry, [1]

(b) two subjects only, [1]

(c) One subject only? [1]

Solution

(i) $x + 5 + 15 + 70 + 10 + x + x - 5 = 170$

$3x + 95 = 170$

$3x = 170 - 95$

$3x = 75$

$x = \frac{75}{3}$

$x = 25$

(ii) (a) **15 learners**

(b) **45 learnes**

(c) **55 learners**

2023.P2.Nov

A survey carried out in a certain district showed that out of 215 teachers, 170 taught Mathematics, 100 taught Additional Mathematics, 41 taught Computer Studies, 65 taught Mathematics and Additional Mathematics, 30 taught Additional Mathematica and Computer Studies, 26 taught Computer Studies and Mathematics and 21 taught all the three subjects.

(i) Illustrate this information on Venn diagram. [2]

(ii) How many teachers taught

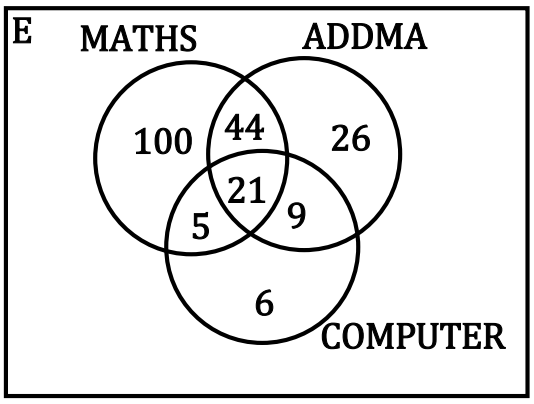
(a) none of the subjects carried out in the survey, [1]

(b) Additional Mathematics or Computer Studies but not Mathematics, [1]

(c) two subjects only? [1]

Solution

(i)



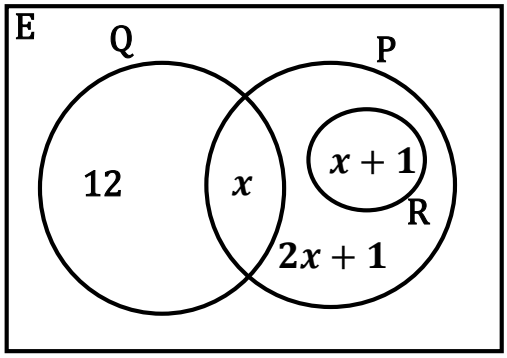
(ii) (a) 4 teachers

(b) 41 teachers

(c) 58 teachers

2024.P2.GCE

The Venn diagram shows the number of elements in sets P, Q and R.



Find

(i) The value of x, such that $n(P) = 18$, [2]
(ii) $n(R)'$, [1]
(iii) $n(R \cup Q)$, [1]
(iv) $n(P \cap Q)'$. [1]

Solution

(i) $\mathbf{n(P) = 18}$

$$x + x + 1 + 2x + 1 = 18$$
$$4x + 2 = 18$$
$$4x = 18 - 2$$
$$4x = 16$$
$$x = \frac{16}{4}$$
$$x = 4$$

(ii) $\mathbf{n(R)' = 12 + 9 + 4}$

$$= 25$$

(iii) $\mathbf{n(R \cup Q) = 12 + 4 + 5}$

$$= 21$$

(iv) $\mathbf{n(P \cap Q)' = 12 + 9 + 5}$

$$= 26$$

ALGEBRA

Objectives

By the end of this topic, leaners should

1. Define Algebra.
2. Simplify algebraic fractions by canceling common factors.
3. Apply the rules of exponents to simplify algebraic expressions.
4. Express two or more fractions as single in simplest forms.

Definition

- Algebra is a branch of mathematics that deals with the study of mathematical symbols, equations, and formulas, and the rules for manipulating these symbols.
- It involves the use of variables, constants, and mathematical operations to solve equations and inequalities.

Algebra typically includes the study of:

1. Variables and expressions: Using letters or symbols to represent unknown values or quantities.
2. Equations and inequalities: Statements that express the equality or inequality of two mathematical expressions.
3. Functions: Relations between variables that assign each input to exactly one output.
4. Graphs: Visual representations of equations and functions.
5. Systems of equations: Sets of two or more equations that are solved simultaneously.

(a) Simplify $3(a+5)-a(a-2)$

(b) Factorise completely $km-6ln+3kn-2lm$

Solution

(a) $3(a+5)-a(a-2)$
$3a+15-a^2+2a$
$3a+2a+15-a^2$
$5a+15-a^2$

(b) $km-6ln+3kn-2lm$
$km+3kn-2lm-6ln$
$k(m+3n)-2l(m+3n)$
$(k-2l)(m+3n)$

2016.p1.GCE

(a) Simplify $2m-4n-3(3n-m)$

(b) Factorise Completely $a^2-2a+7ab-8b+12b^2$

Solution

(a) $2m - 4n - 3(3n - m)$
$2m - 4n - 9n + 3m$
$2m + 3m - 4n - 9n$
$5m - 13n$

(b) $a^2 - 2a + 7ab - 8b + 12b^2$
$a^2 + 12b^2 + 7ab - 2a - 8b$
$(a^2 + 7ab + 12b^2) - 2(a + 4b)$
$(a^2 + 3ab + 4ab + 12b^2) - 2(a + 4b)$
$[a(a + 3b) + 4b(a + 3b)] - 2(a + 4b)$
$(a + 4b)(a + 3b) - 2(a + 4b)$
$(a + 4b)[(a + 3b - 2)]$

2017.p1.Nov.

(a) Simplify $3x - (y - 2x) - 3y$

(b) Factorise completely $ax^2y - 4ay^3$

Solution

(a) $3x - (y - 2x) - 3y$
$3x - y + 2x - 3y$
$3x + 2x - y - 3y$
$5x - 4y$

(b) $ax^2y - 4ay^3$
$ay(x^2 - 4y^2)$
$ay[x^2 - (2y)^2]$
$ay[(x + 2y)(x - 2y)]$

2017.p1.GCE

(a) Simplify $2a - 7b - 2(a - 3b)$

(b) Factorise completely $\frac{x^2}{4y^2} - \frac{1}{9}$

Solution

(a) $2a - 7b - 2(a - 3b)$

$2a - 7b - 2a + 6b$

$2a - 2a + 6b - 7b$

$-b$

(b) $\frac{x^2}{4y^2} - \frac{1}{9}$

$\left(\frac{x}{2y}\right)^2 - \left(\frac{1}{3}\right)^2$

$\left(\frac{x}{2y} + \frac{1}{3}\right)\left(\frac{x}{2y} - \frac{1}{3}\right)$

2018.p1.Oct.

(a) Factorise completely $18 - 2x^2$.

(b) Expand and simplify $(c + 3)^2$.

Solution

(a) $18 - 2x^2$

$2(9 - x^2)$

$2(3^2 - x^2)$

$2[(3 + x)(3 - x)]$

(b) $(c + 3)^2$

$(c + 3)(c + 3)$

$c(c + 3) + 3(c + 3)$

$c^2 + 3c + 3c + 9$

$c^2 + 6c + 9$

2018.p1.GCE

(a) Simplify $4(x+2y)-(3x-8y)$.

(b) Factorise completely $5x^2-5$.

Solution

(a) $4(x+2y)-(3x-8y)$

$4x+8y-3x+8y$

$4x-3x+8y+8y$

$x+16y$

2019.p1.Nov.

(a) Simplify $4x^2+2y^2-xy-6x^2+2xy$

(b) Factorise completely $4ab+6ac-6bd-9cd$

Solution

(a) $4x^2+2y^2-xy-6x^2+2xy$

$4x^2-6x^2+2xy-xy+2y^2$

$-2x^2+xy+2y^2$

$2y^2-2x^2+xy$

(b) $4ab+6ac-6bd-9cd$

$2a(2b+3c)-3d(2b+3c)$

$(2a-3d)(2b+3c)$

2019 p1.GCE

(a) Simplify $4-2(b-a)-1$.

(b) Factorise completely $32x^2-50$.

Solution

(a) $4-2(b-a)-1$

$4-2b+2a-1$

$2a-2b+4-1$

$2a-2b+3$

(b) $5x^2-5$

$5(x^2-1)$

$5(x^2-1^2)$

$5[(x+1)(x-1)]$

(b) $32x^2-50$

$2(16x^2-25)$

2020.p1.Nov

(a) Simplify $2(x+3)-3x(x-2)$

(b) Factorise completely $16x^2-100$

Solution

(a) $2(x+3)-3x(x-2)$

$2x+6-3x^2+6$

$6+6+2x-3x^2$

$12+2x-3x^2$

(b) $16x^2-100$

$4(4x^2-25)$

$4[(2x)^2-5^2]$

$4[(2x+5)(2x-5)]$

2020.p1.GCE

(a) Simplify $2a+(b-a)-2b$

(b) Factorise completely $6ax-4ay-3bx+2by$

Solution

(a) $2a+(b-a)-2b$

$2a+b-a-2b$

$2a-a+b-2b$

$a-b$

(b) $6ax-4ay-3bx+2by$

2021.p1.Nov

(a) Simplify $2y - 3(x - 4) - y$

(b) Factorise completely $3x^3 - 12xy^2$

Solution

(a) $2y - 3(x - 4) - y$

$2y - 3x + 12 - y$

$2y - y - 3x + 12$

$y - 3x + 12$

(b) $3x^3 - 12xy^2$

$3x(x^2 - 4y^2)$

$3x[x^2 - (2y)^2]$

$3x[(x + 2y)(x - 2y)]$

2021.p1.GCE

(a) Simplify $2ab - 2ab^2 - 7ab + a^2b + 5ab^2$

(b) Factorise completely $2x^2 - x - 6$

Solution

(a) $2ab - 2ab^2 - 7ab + a^2b + 5ab^2$

$2ab - 7ab + 5ab^2 - 2ab^2 + a^2b$

$-5ab + 3ab^2 + a^2b$

$3ab^2 + a^2b - 5ab$

(b) $2x^2 - x - 6$

2022.p1.Nov

(a) Simplify $3(4x-5)+2$

(b) Factorise completely $2ax+4ay-3bx-6by$

Solution

(a) $3(4x-5)+2$

$12x-15+2$

$12x+2-15$

$12x-13$

(b) $2ax+4ay-3bx-6by$

$2a(x+2y)-3b(x+2y)$

$(2a-3b)(x+2y)$

2022.p1.GCE

(a) Expand and simplify $(2a+3)(3a-4)$.

(b) Factorise completely $2ab+xy-2ay-bx$.

Solution

(a) $(2a+3)(3a-4)$

$2a(3a-4)+3(3a-4)$

$6a^2-8a+9a-12$

$6a^2+a-12$

(b) $2ab+xy-2ay-bx$

$2ab-2ay-bx+xy$

$2a(b-y)-x(b-y)$

$(2a-x)(b-y)$

2023.p1.Nov

(a) Simplify $2a - 5b(a - b) + ab$

(b) Factorise completely $3x^3 - 27x$

Solution

(a) $2a - 5ab + 5b^2 + ab$

$2a + ab - 5ab + 5b^2$

$2a - 4ab + 5b^2$

(b) $3x^3 - 27x$

$3x(x^2 - 9)$

$3x(x^2 - 3^2)$

$3x[(x + 3)(x - 3)]$

2023.p1.GCE

(a) Simplify $2(x + 3y) - 3(4x - 2)$

(b) Factorise completely $3x^2 - 27$

Solution

(a) $2(x + 3y) - 3(4x - 2)$

$2x + 6y - 12x + 6$

$6y + 6 + 2x - 12x$

$6y + 6 - 10x$

(b) $3x^2 - 27$

$3(x^2 - 9)$

$3(x^2 - 3^2)$

$3[(x + 3)(x - 3)]$

2024.p1.GCE

(a) Simplify $4(x-2)+3(2x-3)$

(b) Factorise completely $7x^2-7$

Solution

(a) $4(x-2)+3(2x-3)$

$4x-8+6x-9$

$4x+6x-8-9$

$10x-17$

(b) $7x^2-7$

$7(x^2-1)$

$7(x^2-1^2)$

$7[(x+1)(x-1)]$

2018.P2.GCE

(a) Simplify $\frac{7st^3}{15u^3v^2} \times \frac{5u^3v}{28s^3t^2}$. [2]

(b) Express $\frac{3}{2x-5} - \frac{4}{x-3}$ as a single fraction in its lowest terms. [3]

Solution

(a) $\frac{7st^3}{15u^3v^2} \times \frac{5u^3v}{28s^3t^2}$

$\frac{35st^3u^3v}{420s^3t^2u^3v^2}$

$\frac{t}{12s^2v}$

(b) $\frac{3}{2x-5} - \frac{4}{x-3}$

$\frac{3(x-3)-4(2x-5)}{(2x-5)(x-3)}$

$\frac{3x-9-8x+20}{(2x-5)(x-3)}$

$\frac{20-9+3x-8x}{(2x-5)(x-3)}$

$$\frac{11-5x}{(2x-5)(x-3)}$$

2019.P2.Nov

(a) Simplify $\frac{6x^2-24}{x-2}$. [2]

(b) Express $\frac{5}{2x-4}-\frac{4}{3x+5}$ as a single fraction in its simplest form. [3]

(c) Simplify $\frac{25p^4}{7q^2}\div\frac{5p^6}{21q^4}\times\frac{p}{15q}$. [2]

Solution

(a)

$$\frac{6x^2-24}{x-2}$$

$$\frac{6(x^2-4)}{x-2}$$

$$\frac{6(x^2-2^2)}{x-2}$$

$$\frac{6[(x+2)(x-2)]}{x-2}$$

$$6(x+2)$$

$$6x+12$$

(b)

$$\frac{5}{2x-4}-\frac{4}{3x+5}$$

$$\frac{5(3x+5)-4(2x-4)}{(2x-4)(3x+5)}$$

$$\frac{15x+25-8x+16}{(2x-4)(3x+5)}$$

(c)

$$\frac{25p^4}{7q^2}\div\frac{5p^6}{21q^4}\times\frac{p}{15q}$$

$$\frac{25p^4}{7q^2}\times\frac{21q^4}{5p^6}\times\frac{p}{15q}$$

$$\frac{525p^5q^4}{525p^6q^3}$$

$$\frac{q}{p}$$

$$\frac{15x-8x+25+16}{(2x-4)(3x+5)}$$

$$\frac{7x+41}{(2x-4)(3x+5)}$$

2020.P2.Nov

(a) Simplify $\frac{2-18p^2}{3p+1}$. [2]

(b) Express $\frac{4}{2x-1}-\frac{1}{3x+2}$ as a single fraction in its lowest terms. [3]

(c) Simplify $\frac{3x^2y}{8xy^3}\div\frac{9x^3}{4y^4}$. [2]

Solution

(a)

$$\frac{2-18p^2}{3p+1}$$

$$\frac{2(1-9p^2)}{1+3p}$$

$$\frac{2(1^2-3^2p^2)}{1+3p}$$

$$\frac{2[1^2-(3p)^2]}{1+3p}$$

$$\frac{2[(1+3p)(1-3p)]}{1+3p}$$

$$2(1-3p)$$

$$2-6p$$

(b)

$$\frac{4}{2x-1}-\frac{1}{3x+2}$$

$$\frac{4(3x+2)-1(2x-1)}{(2x-1)(3x+2)}$$

$$\frac{12x+8-2x+1}{(2x-1)(3x+2)}$$

(c)

$$\frac{3x^2y}{8xy^3}\div\frac{9x^3}{4y^4}$$

$$\frac{3x^2y}{8xy^3}\times\frac{4y^4}{9x^3}$$

$$\frac{12x^2y^5}{72x^4y^3}$$

$$\frac{y^2}{6x^2}$$

$$\frac{12x-2x+8+1}{(2x-1)(3x+2)}$$

$$\frac{10x+9}{(2x-1)(3x+2)}$$

2021.P2.Nov

(a) Simplify $\frac{a-12}{a^2-144}$. [2]

(b) Express $\frac{5}{1-x}-\frac{3}{x-2}$ as a single fraction in its lowest terms. [3]

(c) Simplify $\frac{39x^3}{28y^4}\div\frac{65x^5}{56y^5}$. [2]

Solution

(a) $\frac{a-12}{a^2-144}$

$\frac{a-12}{a^2-12^2}$

$\frac{a-12}{(a+12)(a-12)}$

$\frac{1}{a+12}$

(b) $\frac{5}{1-x}-\frac{3}{x-2}$

$\frac{5(x-2)-4(1-x)}{(1-x)(x-2)}$

$\frac{5x-10-4+4x}{(1-x)(x-2)}$

$\frac{5x+4x-10-4}{(1-x)(x-2)}$

(c) $\frac{39x^3}{28y^4}\div\frac{65x^5}{56y^5}$

$\frac{39x^3}{28y^4}\times\frac{56y^5}{65x^5}$

$\frac{2\,184x^3y^5}{1\,820y^4x^5}$

$\frac{6y}{5x^2}$

$$\frac{9x-14}{(1-x)(x-2)}$$

2022.P2.Nov

(a) Simplify $\frac{2x+14}{2x^2-98}$. [2]

(b) Express $\frac{2}{x+5}-\frac{5}{3x-4}$ as a single fraction in its simplest form. [3]

(c) Simplify $\frac{42xyz}{56}\times\frac{32xy}{40x^2y}$. [2]

Solution

(a)

$$\frac{2x+14}{2x^2-98}$$

$$\frac{2(x+7)}{2(x^2-49)}$$

$$\frac{x+7}{x^2-7^2}$$

$$\frac{x+7}{(x+7)(x-7)}$$

$$\frac{1}{x-7}$$

(b)

$$\frac{2}{x+5}+\frac{5}{3x-4}$$

$$\frac{2(3x-4)+5(x+5)}{(x+5)(3x-4)}$$

$$\frac{6x-8+5x+25}{(x+5)(3x-4)}$$

$$\frac{6x+5x-8+25}{(x+5)(3x-4)}$$

(c)

$$\frac{42xyz}{56}\times\frac{32xy}{40x^2y}$$

$$\frac{1344x^2y^2z}{2240x^2y}$$

$$\frac{3yx}{5}$$

$$\frac{11x+17}{(x+5)(3x-4)}$$

2023.P2.Nov

(a) Express $\frac{2}{3m-2} - \frac{1}{2m-3}$ as a single fraction in its simplest form. [3]

(b) Simplify $\frac{10p^3q^4}{8m^4n^2} \div \frac{25p^4q^3}{4m^4n^2}$. [2]

(c) Simplify $\frac{3d^2-27}{d+3}$. [2]

Solution

(a) $\frac{2}{3m-2} - \frac{1}{2m-3}$

$\frac{2(2m-3)-1(3m-2)}{(3m-2)(2m-3)}$

$\frac{4m-6-3m+2}{(3m-2)(2m-3)}$

$\frac{4m-3m+2-6}{(3m-2)(2m-3)}$

$\frac{m-4}{(3m-2)(2m-3)}$

(b) $\frac{10p^3q^4}{8m^4n^2} \div \frac{25p^4q^3}{4m^4n^2}$

$\frac{10p^3q^4}{8m^4n^2} \times \frac{4m^4n^2}{25p^4q^3}$

$\frac{40m^4n^2p^3q^4}{200m^4n^2p^4q^3}$

$\frac{q}{5p}$

(c) $\frac{3d^2-27}{d+3}$

$\frac{3(d^2-9)}{d+3}$

$\frac{3(d^2-3^2)}{d+3}$

$\frac{3[(d+3)(d-3)]}{d+3}$

$3(d-3)$

$3d-9$

2024.P2.GCE

(a) Express $\frac{2}{x+2} - \frac{5}{3x-1}$ as a single fraction in its simplest form. [3]

(b) Simplify $\frac{10a}{m^2n^2} \div \frac{5a^3}{2m^2n^2}$. [2]

(c) Simplify $\frac{3x^3-6x^2}{x^2-2x}$. [2]

Solution

(a) $\frac{2}{x+2} - \frac{5}{3x-1}$

$\frac{2(3x-1)-5(x+2)}{(x+2)(3x-1)}$

$\frac{6x-2-5x-10}{(x+2)(3x-1)}$

$\frac{6x-5x-2-10}{(x+2)(3x-1)}$

$\frac{x-12}{(x+2)(3x-1)}$

(b) $\frac{10a}{m^2n^2} \div \frac{5a^3}{2m^2n^2}$

$\frac{10a}{m^2n^2} \times \frac{2m^2n^2}{5a^3}$

$\frac{20am^2n^2}{5a^3m^2n^2}$

(c) $\frac{3x^3-6x^2}{x^2-2x}$

$\frac{3x^2(x-2)}{x(x-2)}$

$\frac{3x^2}{x}$

$3x$

$\frac{4}{a^2}$ or $\left(\frac{2}{a}\right)^2$ or $4a^{-2}$

MATRICES

Objectives

By the end of this topic, learners should

1. Define what a matrix is and provide examples.
2. Identify and explain the different types of matrices (row matrix, column matrix, square matrix, diagonal matrix).
3. Understand matrix notation and indexing.
4. Perform basic matrix operations (addition, subtraction, multiplication).
5. Apply matrices to real-world problems.

Definition

- A matrix is a rectangular array of numbers, symbols, or expressions, arranged in rows and columns.

Types of Matrices

1. Row Matrix: A matrix with only one row.
2. Column Matrix: A matrix with only one column.
3. Square Matrix: A matrix with the same number of rows and columns.
4. Diagonal Matrix: A square matrix with non-zero elements only on the main diagonal.
5. Identity Matrix: A square matrix with ones on the main diagonal and zeros elsewhere.
6. Zero Matrix: A matrix with all elements equal to zero.

Matrix Operations

1. Addition: Element-wise addition of corresponding elements.
2. Subtraction: Element-wise subtraction of corresponding elements.
3. Multiplication: Row-column multiplication of corresponding elements.

Matrix Properties

1. Commutativity: Matrix addition is commutative, but matrix multiplication is not.
2. Associativity: Matrix addition and multiplication are associative.
3. Distributivity: Matrix multiplication distributes over matrix addition.

Matrix Applications

1. Linear Algebra: Matrices are used to represent linear transformations and solve systems of linear equations.
2. Computer Graphics: Matrices are used to perform transformations and projections in computer graphics.
3. Data Analysis: Matrices are used to represent and analyze data in statistics and data science.
4. Machine Learning: Matrices are used to represent and manipulate data in machine learning algorithms.

Key Matrix Concepts

1. Matrix Notation: Aij, where A is the matrix name, i is the row number, and j is the column number.
2. Matrix Indexing: Rows are indexed from top to bottom, and columns are indexed from left to right.
3. Matrix Dimensions: The number of rows and columns in a matrix.
4. Matrix Rank: The maximum number of linearly independent rows or columns in a matrix.

The Determinant and Inverse of a 2×2 Matrix

- If matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then its determinant and inverse are calculated as follows:

$$\text{Det}(A) = (a \times d) - (b \times c)$$
$$|A| = ad - bc$$

$$A^{-1} = \frac{1}{|A|}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

2019.p1.Nov

(a) The transpose of matrix A is $(-1 \quad 4 \quad 5)$. Find matrix A.

(b) Given that $(1 \quad 2 \quad 3)\begin{pmatrix} 1 \\ x \\ 5 \end{pmatrix} = (24)$, find the value of x.

Solution

(a) $A^T = (-1 \quad 4 \quad 5)$

$$A = \begin{pmatrix} -1 \\ 4 \\ 5 \end{pmatrix}$$

(b) $(1 \quad 2 \quad 3)\begin{pmatrix} 1 \\ x \\ 5 \end{pmatrix} = (24)$

$$(1 + 2x + 15) = (24)$$
$$(2x + 16) = (24)$$
$$2x + 16 = 24$$
$$2x = 24 - 16$$
$$2x = 8$$
$$x = \frac{8}{2}$$
$$x = 4$$

2019.p1.GCE

Given that $R = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$ and $S = \begin{pmatrix} 1 & 0 & -4 \\ -2 & 1 & 1 \end{pmatrix}$, find

(a) S^T,

(b) RS.

Solution

(a) $S^T = \begin{pmatrix} 1 & -2 \\ 0 & 1 \\ -4 & 1 \end{pmatrix}$

(b) $RS = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}\begin{pmatrix} 1 & 0 & -4 \\ -2 & 1 & 1 \end{pmatrix}$

$= \begin{pmatrix} 2+2 & 0-1 & -8-1 \\ 1-6 & 0+3 & -4+3 \end{pmatrix}$

$= \begin{pmatrix} 4 & -1 & -9 \\ -5 & 3 & -1 \end{pmatrix}$

2020.p1.Nov

Given that $M = \begin{pmatrix} -3 & -1 \\ 4 & 2 \\ 0 & -2 \end{pmatrix}$ and $N = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$, find

(a) M^T,

(b) MN as a single matrix.

Solution

(a) $M^T = \begin{pmatrix} -3 & 4 & 0 \\ -1 & 2 & -2 \end{pmatrix}$

(b) $MN = \begin{pmatrix} -3 & -1 \\ 4 & 2 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \end{pmatrix}$

$= \begin{pmatrix} 6 - 1 \\ -8 + 2 \\ 0 - 2 \end{pmatrix}$

$= \begin{pmatrix} 5 \\ -6 \\ -2 \end{pmatrix}$

2020.p1.GCE

Given that $A = \begin{pmatrix} 2 & 3 \\ 1 & 0 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 0 \\ x & 2 \end{pmatrix}$ and $C = \begin{pmatrix} 7 & 6 \\ -1 & 0 \end{pmatrix}$, find

(a) C^T,

(b) x for which $AB = C$.

Solution

(a) $C^T = \begin{pmatrix} 7 & -1 \\ 6 & 0 \end{pmatrix}$

(b) $AB = C$

$$\begin{pmatrix} 2 & 3 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} -1 & 0 \\ x & 2 \end{pmatrix} = \begin{pmatrix} 7 & 6 \\ -1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} -2+3x & 0+6 \\ -1+0 & 0+0 \end{pmatrix} = \begin{pmatrix} 7 & 6 \\ -1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 3x-2 & 6 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 7 & 6 \\ -1 & 0 \end{pmatrix}$$

$$3x - 2 = 7$$
$$3x = 7 + 2$$
$$3x = 9$$
$$x = \frac{9}{3}$$
$$x = 3$$

2021.p1.Nov

(a) Matrix $M = \begin{pmatrix} -2 & 3 \\ 4 & -2 \end{pmatrix}$. Find the transpose of M.

(b) Matrix $P = \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}$ and $Q = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$. Find PQ.

Solution

(a) $M^T = \begin{pmatrix} -2 & 4 \\ 3 & -2 \end{pmatrix}$

(b) $PQ = \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} -1 \\ 3 \end{pmatrix}$

$= \begin{pmatrix} -2-3 \\ 0+9 \end{pmatrix}$

$= \begin{pmatrix} -5 \\ 9 \end{pmatrix}$

2021.p1.GCE

(a) Given that $A = \begin{pmatrix} 1 & -3 & 5 \\ 2 & 1 & 0 \end{pmatrix}$, find the transpose of A.

(b) Matrix $P = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}$ and $Q = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$. Find PQ as a single matrix.

solution

(a) $A^T = \begin{pmatrix} 1 & 2 \\ -3 & 1 \\ 5 & 0 \end{pmatrix}$

(b) $PQ = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} -2 \\ 3 \end{pmatrix}$

$= \begin{pmatrix} -4-3 \\ -6+12 \end{pmatrix}$

$= \begin{pmatrix} -7 \\ 6 \end{pmatrix}$

2022.p1.Nov

Given that $M = \begin{pmatrix} 1 & 5 \\ 2 & 7 \end{pmatrix}$ and $N = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}$. Find

(a) M^T,

(b) NM.

Solution

(a) $M^T = \begin{pmatrix} 1 & 2 \\ 5 & 7 \end{pmatrix}$

(b) $NM = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}\begin{pmatrix} 1 & 5 \\ 2 & 7 \end{pmatrix}$

$= \begin{pmatrix} 0+2 & 0+7 \\ 2+0 & 10+0 \end{pmatrix}$

$= \begin{pmatrix} 2 & 7 \\ 2 & `0 \end{pmatrix}$

2022.p1.GCE

(a) Given that the matrix $C = \begin{pmatrix} 7 & 3 \\ 1 & 4 \\ 2 & 5 \end{pmatrix}$, find C^T.

(b) Express as a single matrix $(1 \quad 8)\begin{pmatrix} 0 & -1 \\ 2 & 5 \end{pmatrix}$.

Solution

(a) $C^T = \begin{pmatrix} 7 & 1 & 2 \\ 3 & 4 & 5 \end{pmatrix}$

(b) $(1 \quad 8)\begin{pmatrix} 0 & -1 \\ 2 & 5 \end{pmatrix}$

$(0+16 \quad -1+40)$

$(16 \quad 39)$

2023.p1.Nov

Given that $A = (1 \quad 2 \quad 4)$ and $B = \begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix}$. Find

(a) A^T,
(b) AB.

Solution

(a) $A^T = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$

(b) $AB = (1 \quad 2 \quad 4)\begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix}$

$(3 + 0 + 16)$

(19)

2023.p1.GCE

Given that $P = \begin{pmatrix} 4 & -1 & 0 \\ 2 & 0 & 3 \end{pmatrix}$ and $Q = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, find

(a) P^T,
(b) PQ

Solution

(a) $P^T = \begin{pmatrix} 4 & 2 \\ -1 & 0 \\ 0 & 3 \end{pmatrix}$

(b) $PQ = \begin{pmatrix} 4 & -1 & 0 \\ 2 & 0 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

$\begin{pmatrix} 4 - 2 + 0 \\ 2 + 0 + 9 \end{pmatrix}$

$\begin{pmatrix} 2 \\ 11 \end{pmatrix}$

2024.p1.GCE

(a) Given that the matrix $A = (2 \quad -1 \quad 3)$, find A^T.

(b) Given that $A = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and $B = (1 \quad -2)$, find AB

Solution

(a) $A = (2 \quad -1 \quad 3)$

$$A^T = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$$

(b) $AB = \begin{pmatrix} 3 \\ 2 \end{pmatrix}(1 \quad -2)$

$$= \begin{pmatrix} 3 & -6 \\ 2 & -4 \end{pmatrix}$$

2018.P2.GCE

Given that matrix $A = \begin{pmatrix} 2x & 2 \\ 3 & x \end{pmatrix}$,

(i) find the positive value of x for which the determinant of A is 12, [2]

(ii) hence or otherwise, write A^{-1}. [2]

Solution

(i) $(2x \times x) - (2 \times 3) = 12$

$2x^2 - 6 = 12$

$2x^2 = 12 + 6$

$2x^2 = 18$

$x^2 = \frac{18}{2}$

$x^2 = 9$

$x = \pm\sqrt{9}$

$x = \pm 3$

$\therefore x = 3$

(ii) $A^{-1} = \frac{1}{12}\begin{pmatrix} 3 & -2 \\ -3 & 6 \end{pmatrix}$

$= \begin{pmatrix} \frac{3}{12} & -\frac{2}{12} \\ -\frac{3}{12} & \frac{6}{12} \end{pmatrix}$

$= \begin{pmatrix} \frac{1}{4} & -\frac{1}{6} \\ -\frac{1}{4} & \frac{1}{2} \end{pmatrix}$

2019.P2.Nov

Given that matrix A $= \begin{pmatrix} 6 & x \\ 2 & 3 \end{pmatrix}$,

(i) find the value of x for which the determinant of A is 36, [2]

(ii) hence, find the inverse of A. [2]

Solution

(i) $(6 \times 3) - (2 \times x) = 36$

$18 - 2x = 36$

$-2x = 36 - 18$

$-2x = 18$

$x = \frac{18}{-2}$

$x = -9$

(ii) $A^{-1} = \frac{1}{36}\begin{pmatrix} 3 & 9 \\ -2 & 6 \end{pmatrix}$

$= \begin{pmatrix} \frac{3}{36} & \frac{9}{36} \\ -\frac{2}{36} & \frac{6}{36} \end{pmatrix}$

$= \begin{pmatrix} \frac{1}{12} & \frac{1}{4} \\ -\frac{1}{18} & \frac{1}{6} \end{pmatrix}$

2020.P2.Nov

Given that matrix A $= \begin{pmatrix} 5 & 2 \\ 2 & x \end{pmatrix}$, find the

(i) value of x for which A has no inverse, [2]

(ii) inverse of A if $x = 1$. [2]

Solution

(i) $(5 \times x) - (2 \times 4) = 0$

$5x - 4 = 0$

$5x = 0 + 4$

$5x = 4$

$x = \frac{4}{5}$

(ii) $A^{-1} = \frac{1}{1}\begin{pmatrix} 1 & -2 \\ -2 & 5 \end{pmatrix}$

$= \begin{pmatrix} \frac{1}{1} & \frac{-2}{1} \\ \frac{-2}{1} & \frac{5}{1} \end{pmatrix}$

$= \begin{pmatrix} 1 & -2 \\ -2 & 5 \end{pmatrix}$

2021.P2.Nov

Given that matrix $P = \begin{pmatrix} 1 & 2 \\ 2 & -x \end{pmatrix}$,

(i) find the value of x for which the determinant of P is -3, [2]

(ii) hence, find the inverse of P. [2]

Solution

(i) $(-x \times 1) - (2 \times 2) = -3$

$-x - 4 = -3$

$-x = -3 + 4$

$-x = -1$

$-x = \frac{1}{-1}$

$x = -1$

(ii) $P^{-1} = \frac{1}{-3}\begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix}$

$= \begin{pmatrix} \frac{1}{-3} & \frac{-2}{-3} \\ \frac{-2}{-3} & \frac{1}{-3} \end{pmatrix}$

$= \begin{pmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{pmatrix}$

2022.P2.Nov

Given that the determinant of matrix Q $= \begin{pmatrix} 2b-1 & 4 \\ -3b & -5 \end{pmatrix}$ is 15, find the

(i) value of b, [2]
(ii) inverse of matrix Q. [2]

Solution

(i) $-5(2b-1)-(-3b\times 4)=15$

$-10b+5+12b=15$

$2b=15-5$

$2b=10$

$b=\frac{10}{2}$

$b=5$

(ii) $Q^{-1}=\frac{1}{15}\begin{pmatrix} -5 & -4 \\ 15 & 9 \end{pmatrix}$

$=\begin{pmatrix} \frac{-5}{15} & \frac{-4}{15} \\ \frac{15}{15} & \frac{9}{15} \end{pmatrix}$

$=\begin{pmatrix} -\frac{1}{3} & -\frac{4}{15} \\ 1 & \frac{3}{5} \end{pmatrix}$

2023.P2.Nov

Given that matrix K = $\begin{pmatrix} 8 & 9x \\ 1 & x \end{pmatrix}$, find the

(i) value of x for which the determinant of K is 2, [2]

(ii) inverse of K. [2]

Solution

(i) $(x \times 8) - (9x \times 1) = 2$

$8x - 9x = 2$

$-x = 2$

$-x = \frac{2}{-1}$

$x = -2$

(ii) $K^{-1} = \frac{1}{2}\begin{pmatrix} -2 & 18 \\ -1 & 8 \end{pmatrix}$

$= \begin{pmatrix} \frac{-2}{2} & \frac{18}{2} \\ \frac{-1}{2} & \frac{8}{2} \end{pmatrix}$

$= \begin{pmatrix} -1 & \frac{2}{3} \\ -\frac{1}{2} & 4 \end{pmatrix}$

2024.P2.GCE

Given that matrix B $= \begin{pmatrix} 4 & x \\ -1 & -x3 \end{pmatrix}$,

(i) find the value of x for which the determinant of B is 22, [2]

(ii) hence, find the inverse of B. [2]

Solution

(i) $4(-3x) - (-1 \times x) = 22$

$-12x - (-x) = 22$

$-12x + x = 22$

$-11x = 22$

$x = \frac{22}{-11}$

$x = -2$

(ii) $B = \begin{pmatrix} 4 & -2 \\ -1 & 6 \end{pmatrix}$

$B^{-1} = \frac{1}{22}\begin{pmatrix} 6 & 2 \\ 1 & 4 \end{pmatrix}$

$= \begin{pmatrix} \frac{6}{22} & \frac{2}{22} \\ \frac{1}{22} & \frac{4}{22} \end{pmatrix}$

$= \begin{pmatrix} \frac{3}{11} & \frac{1}{11} \\ \frac{1}{22} & \frac{2}{11} \end{pmatrix}$

INDEX NOTATION

- Index notation is a method of representing numbers and letters that have been multiplied by themselves multiple times.
- An index number is defined as the number which is raised to the power.
- The power says how many times the number is to be used in multiplication.
- Generally, it is represented as a small number to the right side and above the base number.
- When two numbers with the same base are multiplied, their powers get added.
- When two numbers with the same base are divided, their powers get subtracted.
- If any term with power is raised to the exponent or power, the exponents or powers are multiplied together.
- Any negative powers can be represented in a fractional form.
- The exponent or index given in a fraction form can be represented as the radical form.
- Power of 10 is a unique way of writing large numbers or smaller numbers.
- Instead of using so many zeroes, you can show how many powers of the 10 will make that many zeroes.
- For example, 6 000 in the power of 10 can be written as 6×10^3.
- Power of 10 is extremely used by Scientists and Engineers as they deal with the numbers that include large numbers of zeroes.

Laws of Indices

1 $a^m \times a^n = a^{m+n}$

2 $a^m \times b^m = (ab)^m$

3 $(a^m)^n = a^{m \times n}$

4 $a^m \div a^n = a^{m-n}$

5 $a^m \div b^m = \left(\frac{a}{b}\right)^m$

6 $a^1 = a$

7 $a^0 = 1$

8 $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$

9 $a^{-m} = \frac{1}{a^m}$

10 $\left(\frac{a}{b}\right)^{-m} = \left(\frac{b}{a}\right)^m$

11 $a^{\frac{n}{m}} = \left(\sqrt[n]{a}\right)^m$

2016.p1.Specimen

(a) Evaluate $\left(\sqrt[3]{64}\right)^2$

(b) Solve the equation $2^{2m-3} = 8^m$

Solution

(a) $\left(\sqrt[3]{64}\right)^2$

$\left(\sqrt[3]{4^3}\right)^2$

$(4)^2$

4×4

16

(b) $2^{2m-3} = 8^m$

$2^{2m-3} = (2^3)^m$

$2^{2m-3} = 2^{3m}$

$2m - 3 = 3m$

$2m + 3m = 3$

$5m = 3$

$m = \frac{3}{5}$

2016.p1.Oct

(a) Evaluate $8^{\frac{2}{3}}$

(b) Evaluate $5^0 + 5^1$

Solution

(b) $8^{\frac{2}{3}}$

$\left(\sqrt[3]{8}\right)^2$

2^2

2×2

4

(a) $5^0 + 5^1$

$1 + 5$

6

2016.p1.GCE

(a) Evaluate $3^2 + 2^3 \times 2^0$

(b) It is given that $10^x = 3$ and $10^y = 9$. What is the value of 10^{y-x}?

Solution

(a) $3^2 + 2^3 \times 2^0$

$(3 \times 3) + (2 \times 2 \times 2) \times 1$

$9 + 8 \times 1$

$9 + 8$

17

(b) 10^{y-x}

$10^y \div 10^x$

$9 \div 3$

3

2017.p1.Nov

(a) Evaluate $\left(\frac{81}{16}\right)^{-\frac{1}{4}} + \left(\frac{81}{16}\right)^0$

(b) Solve the equation $10^x = 0.0001$

Solution

(a) $\left(\frac{81}{16}\right)^{-\frac{1}{4}} + \left(\frac{81}{16}\right)^0$

$\left(\frac{16}{81}\right)^{\frac{1}{4}} + 1$

$\left(\sqrt[4]{\frac{16}{81}}\right)^1 + 1$

$\frac{2}{3} + 1$

$1\frac{2}{3}$

(b) $10^x = 0.0001$

$10^x = 10^{-4}$

$x = -4$

2017.p1.GCE

(a) Solve the equation $25^x = 5$

(b) Given that $x = 3.2 \times 3^4$ and $y = 4 \times 3^2$, evaluate $\frac{x}{y}$.

Solution

(b) $25^x = 5$

$(5^2)^x = 5^1$

$5^{2x} = 5^1$

$2x = 1$

$x = \frac{1}{2}$

(a) $\frac{x}{y}$

$\frac{3.2\times3^4}{4\times3^2}$

$\frac{3.2\times3\times3\times3\times3}{4\times3\times3}$

$\frac{28.8}{4}$

7.2

2018.p1.Nov

(a) Find the value of $x^0 \times 2^3$

(b) Solve the equation $x^{\frac{1}{3}} = 4$

Solution

(a) $x^0 \times 2^3$

$1 \times 2 \times 2 \times 2$

1×8

8

(b) $x^{\frac{1}{3}} = 4$

$\left(x^{\frac{1}{3}}\right)^3 = 4^3$

$x = 4 \times 4 \times 4$

$x = 64$

2018.p1.GCE

(a) Evaluate $27^{\frac{2}{3}}$

(b) Solve the equation $2^{2x-1} = 16^{-2x}$

Solution

(a) $27^{\frac{2}{3}}$

$(3^3)^{\frac{2}{3}}$

$3^{\frac{6}{3}}$

3^2

3×3

9

(b) $22^{2x-1} = 16^{-2x}$

$2^{2x-1} = (2^4)^{-2x}$

$2^{2x-1} = 2^{-8x}$

$2x - 1 = -8x$

$2x + 8x = 1$

$10x = 1$

$x = \frac{1}{10}$

2019.p1.Nov

(a) Evaluate $(-1)^0 \times 2^3$

(b) Solve the equation $8^x = 128$

Solution

(a) $(-1)^0 \times 2^3$

1×8

8

(b) $8^x = 128$

$(2^3)^x = 2^7$

$2^{3x} = 2^7$

$3x = 7$

$x = \frac{7}{3}$

$x = 2\frac{1}{3}$

2019.p1.GCE

(a) Evaluate $\left(\sqrt[4]{81}\right)^3$

(b) Evaluate $3^0 \times 3^3 + 3^1$

Solution

(a) $\left(\sqrt[4]{81}\right)^3$

3^3

27

(b) $3^0 \times 3^3 + 3^1$

$1 \times 27 + 3$

$27 + 3$

30

2020.p1.Nov

(a) Evaluate $\left(\sqrt[3]{27}\right)^2$

(b) Solve the equation $3^{x+3} = 27^5$

Solution

(a) $\left(\sqrt[3]{27}\right)^2$

3^2

9

(b) $3^{x+3} = 27^5$

$3^{x+3} = (3^3)^5$

$3^{x+3} = 3^{15}$

$x + 3 = 15$

$x = 15 - 3$

$x = 12$

2020.p1.GCE

(a) Evaluate $\left(\frac{64}{125}\right)^{-\frac{1}{3}}$

(b) Given that $8^{x-1} = 16$, find the value of x.

Solution

(a)
$$\left(\frac{64}{125}\right)^{-\frac{1}{3}}$$
$$\left(\frac{125}{64}\right)^{\frac{1}{3}}$$
$$\left(\sqrt[3]{\frac{125}{64}}\right)^{1}$$
$$\frac{5}{4}$$
$$1\frac{1}{4}$$

(b)
$$8^{x-1} = 16$$
$$(2^3)^{x-1} = 2^4$$
$$2^{3x-3} = 2^4$$
$$3x - 3 = 4$$
$$3x = 4 + 3$$
$$3x = 7$$
$$x = \frac{7}{3}$$
$$x = 2\frac{1}{3}$$

2021.p1.Nov

(a) Evaluate $-3^2 + 3^2$

(b) Solve the equation $\sqrt{2} = 2^x$

Solution

(a)
$$-3^2 + 3^2$$
$$-9 + 9$$
$$0$$

(b)
$$\sqrt{2} = 2^x$$
$$2^{\frac{1}{2}} = 2^x$$
$$x = \frac{1}{2}$$

2021.p1.GCE

(a) Evaluate $9^{\frac{1}{2}} + 1$

(b) Solve the equation $125^x = \frac{1}{25}$

Solution

(a) $9^{\frac{1}{2}} + 1$

$\sqrt{9} + 1$

$3 + 1$

4

(b) $125^x = \frac{1}{25}$

$(5^3)^x = 25^{-1}$

$5^{3x} = (5^2)^{-1}$

$5^{3x} = 5^{-2}$

$3x = -2$

$x = -\frac{2}{3}$

2022.p1.Nov

(a) Evaluate $\left(\frac{3}{2}\right)^{-2}$

(b) Solve the equation $2x^3 = 16$

Solution

(a) $\left(\frac{3}{2}\right)^{-2}$

$\left(\frac{2}{3}\right)^{2}$

$\frac{2^2}{3^2}$

$\frac{4}{9}$

(b) $2x^3 = 16$

$x^3 = \frac{16}{2}$

$x^3 = 8$

$x^3 = 2^3$

$x = 2$

2022.p1.GCE

(a) Evaluate $5^3 \times \frac{1}{5^2}$

(b) Solve the equation $x^{\frac{1}{3}} = 9$

Solution

(a) $5^3 \times \frac{1}{5^2}$

$\frac{5^3}{5^2}$

5

(b) $x^{\frac{1}{3}} = 9$

$\left(x^{\frac{1}{3}}\right)^3 = 9^3$

$x = 9 \times 9 \times 9$

$x = 729$

2023.p1.Nov

(a) Evaluate $\left(\frac{8}{27}\right)^{\frac{1}{3}}$

(b) Solve the equation $2^{x+3} = \frac{1}{16}$

Solution

(a) $\left(\frac{8}{27}\right)^{\frac{1}{3}}$

$\left(\sqrt[3]{\frac{8}{27}}\right)^1$

$\left(\frac{2}{3}\right)^1$

$\frac{2}{3}$

(b) $2^{x+3} = \frac{1}{16}$

$2^{x+3} = 16^{-1}$

$2^{x+3} = (2^4)^{-1}$

$2^{x+3} = 2^{-4}$

$x + 3 = -4$

$x = -4 - 3$

$x = -7$

2023.p1.GCE

(a) Evaluate $3^2 + 3^{-1} \times 2^0$

(b) Solve the equation $5^{-3x} = 125$

Solution

(a)
$$3^2 + 3^{-1} \times 2^0$$
$$9 + \frac{1}{3} \times 1$$
$$9 + \frac{1}{3}$$
$$9\frac{1}{3}$$

(b)
$$5^{-3x} = 125$$
$$5^{-3x} = 5^3$$
$$-3x = 3$$
$$x = \frac{-3}{3}$$
$$x = -1$$

2024.p1.GCE

(a) Evaluate $\left(\frac{8}{27}\right)^{\frac{2}{3}}$

(b) Solve the equation $81 - 27^x = 0$

Solution

(a)
$$\left(\frac{8}{27}\right)^{\frac{2}{3}}$$
$$\left(\sqrt[3]{\frac{8}{27}}\right)^2$$
$$\left(\frac{2}{3}\right)^2$$
$$\frac{2^2}{3^2}$$
$$\frac{4}{9}$$

(b)
$$81 - 27^x = 0$$
$$27^x = 81$$
$$(3^3)^x = 3^4$$
$$3^{3x} = 3^4$$
$$3x = 4$$
$$x = \frac{4}{3}$$
$$x = 1\frac{1}{3}$$

SIMILARITY AND CONGRUENCE

- Similarity is one of the areas that were advanced by the Greek Mathematician by the name of Euclid.
- According to Euclid, two geometric figures that have the same shape are said to be similar, despite their difference in size.
- The word congruency comes from the Latin word 'congruence' which means 'to come together, fit in or fall upon.'
- However, in geometry, figures are congruent if they have exactly the same shape and size.
- When looking at yourself in the mirror, your image is of the same shape and size as the object.

2016.p1.Oct.

The base areas of two containers that are geometrically similar are 80cm^2 and 180cm^2, respectively. If the capacity of the larger container is 54 litres, calculate the capacity of the smaller one.

Solution

$$k^2 = 80 : 180$$

$$k^2 = 4 : 9$$

$$k = \sqrt{4} : \sqrt{9}$$

$$k = 2 : 3$$

$$k^3 = 2^3 : 3^3$$

$$k^3 = 8 : 27$$

$$8 : 27 = x : 54$$

$$\frac{8}{27} = \frac{x}{54}$$

$$27x = 8 \times 54$$

$$x = \frac{8 \times 54}{27}$$

$$x = 8 \times 2$$

$$x = 16\ litres$$

2017.p1.Nov.

The ratio of the surface areas of two cubes is 16 : 25. What is the volume of the smaller cube, if the volume of the bigger cube is 500cm³?

Solution

$$k^2 = 16 : 25$$

$$k = \sqrt{16} : \sqrt{25}$$

$$k = 4 : 5$$

$$k^3 = 4^3 : 5^3$$

$$k^3 = 64 : 125$$

$$64 : 125 = x : 500$$

$$\frac{64}{125} = \frac{x}{500}$$

$$125x = 64 \times 500$$

$$x = \frac{64 \times 500}{125}$$

$$x = 64 \times 4$$

$$x = 256cm^3$$

2017.p1.GCE

Two tins are geometrically similar. If the ratio of their volumes is 27 : 64, find the ratio of their curved surface areas.

Solution

$$k^3 = 27 : 64$$

$$k = \sqrt[3]{27} : \sqrt[3]{64}$$

$$k = 3 : 4$$

$$k^2 = 3^2 : 4^2$$

$$k^2 = 9 : 16$$

2018.p1.Nov.

A cylinder with diameter 2cm has a volume of 5cm^3. Find the volume of a similar cylinder with diameter 8cm.

Solution

$$k = 2 : 8$$

$$k = 1 : 4$$

$$k^3 = 1^3 : 4^3$$

$$k^3 = 1 : 64$$

$$1 : 64 = 5 : x$$

$$\frac{1}{64} = \frac{5}{x}$$

$$x = 64 \times 5$$

$$x = 320cm^3$$

2018.P1.GCE

The scale of a map is 1 : 20 000. The actual area of residential plots is 60km^2. Calculate the area of residential plots on the map in square centimeters.

Solution

$$k = 1 : 20\,000$$

$$k^2 = (1 : 20\,000 \div 100 \div 1\,000)^2$$

$$k^2 = 1 : 0.04$$

$$1 : 0.04 = x : 60$$

$$\frac{1}{0.04} = \frac{x}{60}$$

$$x = \frac{60}{0.04}$$

$$x = 1\,500cm^2$$

2019.p1.Nov.

The heights of two similar cylinders are 4cm and 6cm. If the volume of the smaller cylinder is 48cm^3, find the volume of the larger cylinder.

Solution

$$k = 4 : 6$$

$$k = 2 : 3$$

$$k^3 = 2^3 : 3^3$$

$$k^3 = 8 : 27$$

$$8 : 27 = 48 : x$$

$$\frac{8}{27} = \frac{48}{x}$$

$$8x = 27 \times 48$$

$$x = \frac{27 \times 48}{8}$$

$$x = \frac{3 \times 3 \times 3 \times 2 \times 2 \times 2 \times 2 \times 3}{2 \times 2 \times 2}$$

$$x = 3 \times 3 \times 3 \times 3 \times 2$$

$$x = 162cm^3$$

2019.p1.GCE

The ratio of the volumes of two similar solids is 64 : 27. The surface area of the smaller solid is 180cm^2. What is the surface area of the bigger solid?

Solution

$$k^3 = 64 : 27$$

$$k = \sqrt[3]{64} : \sqrt[3]{27}$$

$$k = 4 : 3$$

$$k^2 = 4^2 : 3^2$$

$$k^2 = 16 : 9$$

$$16 : 9 = x : 180$$

$$\frac{16}{9} = \frac{x}{180}$$

$$9x = 16 \times 180$$

$$x = \frac{16 \times 180}{9}$$

$$x = 16 \times 20$$

$$x = 320cm^2$$

2020.P1.Nov

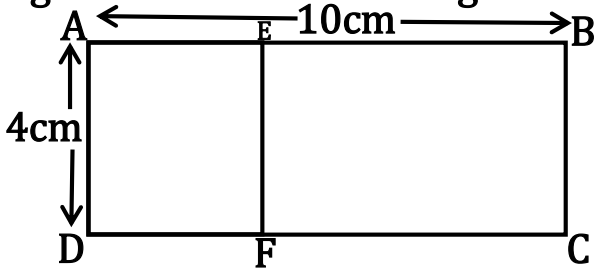
A plan of a play boat is drawn to a scale of 1 :20. If the actual area of the base of the boat is 0.64m^2, calculate its area on the map in square centimetres.

Solution

$$k = 1 : 20$$
$$k^2 = (1 : 20 \div 100)^2$$
$$k^2 = 1 : \ 0.04$$
$$1 : 0.04 = x : 0.64$$
$$\frac{1}{0.04} = \frac{x}{0.64}$$
$$x = \frac{0.64}{0.04}$$
$$x = 16cm^2$$

2020.P1.GCE

Rectangle ABCD and DAEF are geometrically similar.



Given that AB = 10cm and AD = 4cm, calculate the area of rectangle DAEF.

Solution

$$k = 4 : 10$$
$$k = 2 : 5$$
$$k^2 = 2^2 : 5^2$$
$$k^2 = 4 : 25$$
$$\text{Area} = l \times b$$
$$= 4 \times 10$$
$$= 40cm^2$$
$$4 : 25 = x : 40$$
$$\frac{4}{25} = \frac{x}{40}$$
$$25x = 40 \times 4$$
$$x = \frac{40 \times 4}{25}$$
$$x = \frac{32}{5}$$
$$x = 6.4cm^2$$

2021.P1.Nov

The areas of two similar cylinders are $64cm^2$ and $36cm^2$ respectively. If the height of the larger cylinder is 30cm, find the height of the smaller cylinder.

Solution

$$k^2 = 64 : 36$$

$$k = \sqrt{64} : \sqrt{36}$$

$$k = 8 : 6$$

$$k = 4 : 3$$

$$4 : 3 = 30 : h$$

$$\frac{4}{3} = \frac{30}{h}$$

$$4h = 30 \times 3$$

$$h = \frac{30 \times 3}{4}$$

$$h = 22\frac{1}{2}cm$$

$$h = 22.5cm$$

2021.P1.GCE

The areas of the faces of cubes A and B are $9cm^2$ and $25cm^2$ respectively. Find the ratio of their volumes.

Solution

$$k^2 = 9 : 25$$

$$k = \sqrt{9} : \sqrt{25}$$

$$k = 3 : 5$$

$$k^3 = 3^3 : 5^3$$

$$k^3 = 27cm^3 : 125cm^3$$

2022.P1.Nov

Two similar solids P and Q have volumes 80cm^3 and 270cm^3 respectively. The height of the smaller solid is 8cm. Find the height of the larger solid.

Solution

$$k^3 = 80 : 270$$

$$k^3 = 8 : 27$$

$$k = \sqrt[3]{8} : \sqrt[3]{27}$$

$$k = 2 : 3$$

$$2 : 3 = 8 : \mathrm{H}$$

$$\frac{2}{3} = \frac{8}{\mathrm{H}}$$

$$2\mathrm{H} = 8 \times 3$$

$$\mathrm{H} = \frac{8 \times 3}{2}$$

$$\mathrm{H} = 12\mathrm{cm}$$

2022.P1.GCE

Two cylinders A and B are geometrically similar and the ratio of their heights is $m : 4$ respectively. Given that the areas of A and B are 144cm^2 and 256cm^2 respectively, find the value of m.

Solution

$$k^2 = 144 : 256$$

$$k = \sqrt{144} : \sqrt{256}$$

$$k = 12 : 16$$

$$k = 3 : 4$$

$$3 : 4 = m : 4$$

$$m = 3cm$$

2023.P1.Nov.

The areas of two similar shapes are in the ratio 25 : 36. If the length of the smaller shape is 2cm, find the length of the larger shape.

Solution

$$K^2 = 25 : 36$$

$$K = \sqrt{25} : \sqrt{36}$$

$$K = 5 : 6$$

$$5 : 6 = 2 : l$$

$$\frac{5}{6} = \frac{2}{l}$$

$$5l = 12$$

$$l = \frac{12}{5}$$

$$l = 2.4cm$$

2023.P1GCE

Two cylindrical tins are geometrically similar. The radius of the base of the smaller tin is 6cm and that of the bigger tin is 9cm. given that the volume of the bigger tin is 108cm^3, find the volume of the smaller tin.

Solution

$$k = 6 : 9$$

$$k = 2 : 3$$

$$k^3 = 2^3 : 3^3$$

$$k^3 = 8 : 27$$

$$8 : 27 = v : 108$$

$$\frac{8}{27} = \frac{v}{108}$$

$$27v = 108 \times 8$$

$$v = \frac{108 \times 8}{27}$$

$$x = \frac{3 \times 3 \times 3 \times 2 \times 2 \times 2 \times 2 \times 2}{3 \times 3 \times 3}$$

$$x = 2 \times 2 \times 2 \times 2 \times 2$$

$$x = 32cm^3$$

2024.P1GCE

The ratio of the areas of two similar solids is 9 : 25. If the volume of the bigger solid is 375cm³, what is the volume of the smaller solid?

Solution

$$K^2 = 9 : 25$$
$$K^2 = \sqrt{9} : \sqrt{25}$$
$$K = 3 : 5$$

$$K^3 = 3^3 : 5^3$$
$$K^3 = 27 : 125$$

$$27 : 125 = x : 375$$

$$\frac{27}{125} = \frac{x}{375}$$

$$125x = 27 \times 375$$

$$x = \frac{27 \times 375}{125}$$

$$x = 27 \times 3$$

$$x = 81cm^3$$

SOCIAL AND COMMERCIAL ARITHMETICS

- Social and Commercial Arithmetic refers to the practical application of mathematical concepts and techniques to real-life situations involving personal finance, business, commerce, and social transactions.

Shares

- A share represents ownership in a company.
- The following are the types of share:
  - Ordinary shares (common stock)
  - Preference shares (preferred stock)
  - Cumulative preference shares
  - Non-cumulative preference shares
- The following are the features of shares:
  - Voting rights
  - Dividend entitlement
  - Capital appreciation
  - Transferability

Dividends

- A dividend is a portion of company profits distributed to shareholders.
- The following are the types of dividends:
  - Interim dividend
  - Final dividend
  - Special dividend
- Dividend per share (DPS) $= \frac{Total\ dividend}{Total\ number\ of\ shares}$
- Dividend yield $= \frac{DPS}{Market\ price\ per\ share} \times 100$

- Return on investment (ROI) $= \frac{Dividend + Capital\ gains}{Investment}$

Key Concepts:

1. Face value (nominal value): The original price of a share.
2. Market value: The current price of a share.
3. Capital gains: Profits from selling shares.
4. Dividend cover: The number of times dividend payments are covered by earnings.

2018.P1.Oct.

Floyd had 125 shares in a company. At the end of a financial year, the company declared a dividend of K3.00 per share. How much was due to Floyd?

Solution

125 shares $\longrightarrow x$

1 share $\longrightarrow$ K3.00

$\frac{125}{1} = \frac{x}{3}$

$x = 3 \times 125$

$= \text{K}375.00$

2019.P1.Nov.

A company declared a dividend of K1.50 per share. Musalala has 600 shares in the company. How much will she get?

Solution

$$1 \text{ share} \longrightarrow \text{K}1.50$$

$$600 \text{ shares} \longrightarrow x$$

$$\frac{1}{600} = \frac{1.50}{x}$$

$$x = 600 \times 1.50$$

$$= \text{K}900.00$$

2020.P1.Nov.

A woman has 500 company shares with a nominal value of K70.00 each, which she bought at K30.00 each. How much did she pay for the shares?

Solution

$$1 \text{ share} \longrightarrow \text{K}30.00$$

$$500 \text{ shares} \longrightarrow x$$

$$\frac{1}{500} = \frac{30}{x}$$

$$x = 500 \times 30$$

$$= \text{K}15\,000.00$$

2021.P1.Nov.

A certain company sells shares at K1.30 per share. A businesswoman had K9 100.00 to buy shares. How many shares did she get?

Solution

$$1 \text{ share} \longrightarrow \text{K}1.30$$

$$x \longrightarrow \text{K}9\,100$$

$$\frac{1}{x} = \frac{1.30}{9\,100}$$

$$1.30x = 9\,100$$

$$x = \frac{9\,100}{1.30}$$

$$x = \frac{9\,100 \times 100}{1.30 \times 100}$$

$$x = \frac{910\,000}{130}$$

$$x = 7\,000\ shares$$

2022.P1.Nov.

A company declared a dividend of K2.75 per share. A businessman invested 560 shares in the company. How much dividend did he get?

Solution

$$1 \text{ share} \longrightarrow \text{K}2.75$$

$$560 \text{ shares} \longrightarrow x$$

$$\frac{1}{560} = \frac{2.75}{x}$$

$$x = 560 \times 2.75$$

$$= \text{K}1\,540.00$$

2023.P1.Nov.

A sugar company paid K362 400.00 as dividend for 1 200 shares. A man had 20 shares in the company. Calculate the amount paid to him.

Solution

$$120 \text{ shares} \longrightarrow \text{K}362\,400$$

$$20 \text{ shares} \longrightarrow x$$

$$\frac{120}{20} = \frac{362\,400}{x}$$

$$x = \frac{362\,400}{6}$$

$$= \text{K}60\,400.00$$

2024.P1.GCE.

A businessman got 800 shares worth K7 200.00. What was the cost of each share?

Solution

$$1 \text{ share} \longrightarrow x$$

$$800 \text{ shares} \longrightarrow 7\,200$$

$$\frac{1}{800} = \frac{x}{7\,200}$$

$$800x = 7\,200$$

$$x = \frac{7\,200}{800}$$

$$= \text{K}9.00 \text{ per share}$$

SYMMETRY

- Symmetry refers to the quality of being unchanged by a transformation, such as reflection, rotation, or translation.

Types of Symmetry

1. Line Symmetry: A figure has line symmetry if it can be divided into two identical parts by a line.
2. Rotational Symmetry: A figure has rotational symmetry if it looks the same after a rotation of 90°, 180°, 270°, or 360°.
3. Reflection Symmetry: A figure has reflection symmetry if it can be reflected over a line to produce an identical image.
4. Translation Symmetry: A figure has translation symmetry if it can be translated (moved) to produce an identical image.

Examples of Symmetry

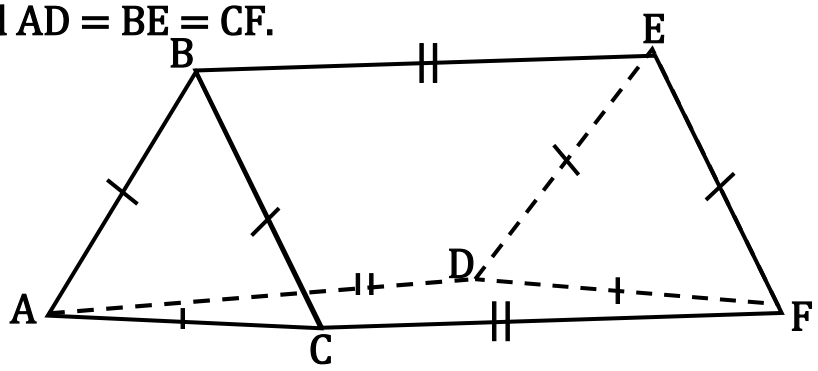
1. Bilateral Symmetry: A butterfly has bilateral symmetry because its left and right wings are identical.
2. Radial Symmetry: A starfish has radial symmetry because its body parts are arranged around a central axis.
3. Rotational Symmetry: A square has rotational symmetry because it looks the same after a rotation of 90°.

Properties of Symmetry

1. Identity: A figure remains unchanged under the identity transformation.
2. Inverse: For every transformation, there is an inverse transformation that "reverses" the original transformation.
3. Composition: The composition of two or more transformations is also a transformation.

2018.P1.Oct

The diagram below shows a triangular prism with AB = AC = DE = EF = DF and AD = BE = CF.



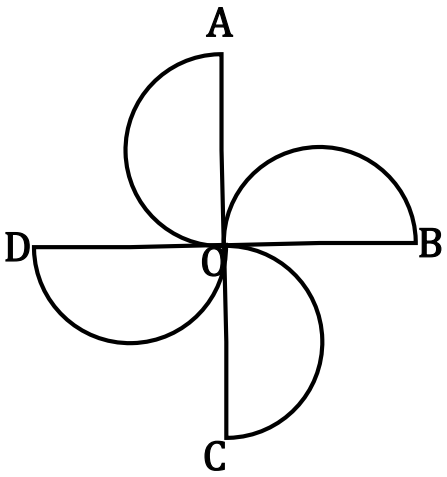
State the number of planes of symmetry of the prism.

Solution

It has 4 planes of symmetry.

2019.P1.Nov

The diagram below shows a plane figure made up of congruent semi circles. Angle AOB = 90°.



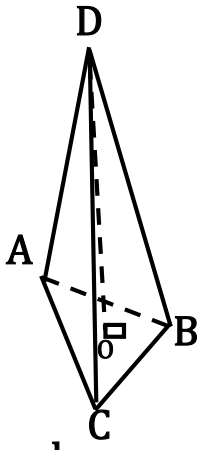
Describe fully the symmetry of the figure.

Solution

It is a rotational symmetry of order 4 about 90°.

2020.P1.Nov.

The diagram below is a right triangular pyramid ABCD with AB = BC = CA.



What is the order of rotational symmetry of the pyramid about OD?

Solution

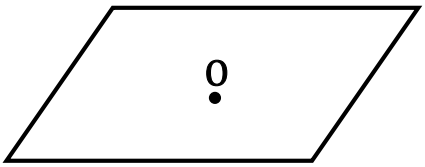
A right triangular pyramid has an order of rotational symmetry of 3, because a right triangular pyramid has a triangular base and three rectangular faces meeting at the apex.

Rotating the pyramid by $120° \; or \; \frac{360°}{3}$ around its central axis (through the apex and centroid of the base) results in the same orientation.

This means the pyramid looks identical after a 120° rotation, demonstrating 3-fold rotational symmetry.

2021.P1.Nov.

The figure below shows a parallelogram with centre O.



Describe fully the symmetry of the parallelogram about the centre O.

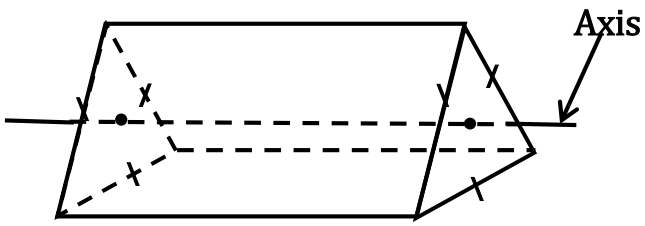
Solution

A parallelogram has rotational symmetry of order 2 about its center O.

Description: When a parallelogram is rotated 180° about its center O, it coincides with its original position.

2022.P1.Nov.

The diagram below shows a regular triangular prism.



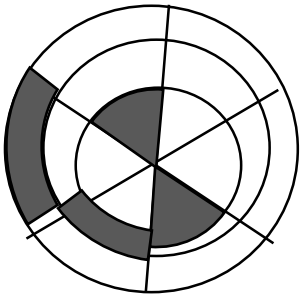
Describe the symmetry of the prism.

Solution

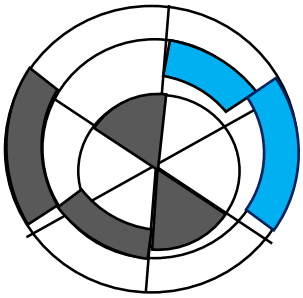
It is a rotational symmetry of order 4 about 90°.

2023.P1.Nov.

Shade two more sections of the diagram below so that it has a rotational symmetry of order 2.

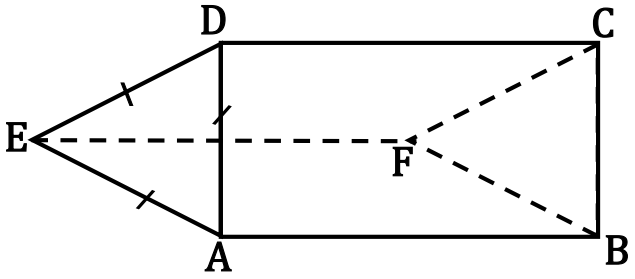


Solution



2024.P1.GCE

The diagram below shows a triangular prism ABCDEF in which AE = AD = DE.



State the number of planes of symmetry for this triangular prism.

Solution

It has 4 planes of symmetry.

COMPUTERS AND CALCULATORS

Computers and calculators are essential tools in mathematics, enabling us to perform calculations, visualize data, and explore mathematical concepts.

Pseudocodes and Flowcharts

- Flowcharts and pseudocodes are essential tools in programming and problem-solving.
- They help to visualize and communicate the logic of a program or algorithm.

Flowcharts

- A flowchart is a graphical representation of a program or algorithm.
- It consists of symbols, such as rectangles, diamonds, parallelograms, ovals and arrows, that represent different stages of the program.

Types of Flowchart Symbols

1. Parallelogram: Represents the displaying, entering, outputting, inputting or printing of data.
2. Rectangle: Represents a process or action, calculations involving formulas.
3. Diamond: Represents a decision or condition.
4. Arrow: Represents the flow of control.
5. Oval: Represents the start or end of a program.

How to Draw a Flowchart

1. Start with a clear problem statement.
2. Identify the inputs, processing, and outputs.
3. Use symbols to represent each stage of the program.
4. Arrange the symbols in a logical order.
5. Use arrows to indicate the flow of control.

Pseudocodes

- A pseudocode is a high-level representation of a program or algorithm.
- It uses a combination of natural language and programming-like syntax to describe the logic of the program.

Advantages of Pseudocodes

1. Easy to understand and write.
2. Platform-independent.
3. Helps to identify errors and bugs.

How to Write a Pseudocode

1. Start with a clear problem statement.
2. Identify the inputs, processing, and outputs.
3. Use a combination of natural language and programming-like syntax.
4. Use indentation to indicate the structure of the program.

2018.P1.Oct

In the answer space below, is an incomplete program in pseudo code to calculate the circumference of a circle of latitude $x°$, given the radius R of the earth. Complete the program by filling in the blank spaces with appropriate statements.

```
Begin
    Enter ……..
    C = ……..
    Output C
End.
```

Solution

```
Begin
    Enter R, x°
    C = 2 * π * R * cos x°
    Output C
End.
```

2019.P1.Nov.

In the answer space below is an incomplete program written in pseudocode for calculating the volume V of a cuboid, given the length, l, base, b, and the height, h. Complete the program by filling in the blank spaces with appropriate statements.

```
Begin
    Enter ……..
    V = ……..
    Output V
End.
```

Solution

```
Begin
    Enter l, b, h
    V = l * b * h
    Output V
End.
```

2020.P1.Nov

In the answer space below, is an incomplete pseudocode for a programme for calculating the curved surface area of a cone. Complete the pseudocode by filling in the blank spaces.

Start

Enter .....

S = ...

Output S

Stop.

Solution

Start

Enter r, l

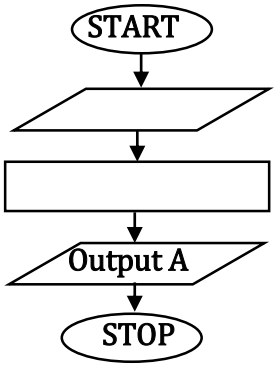
S = $\pi * r * l$

Output S

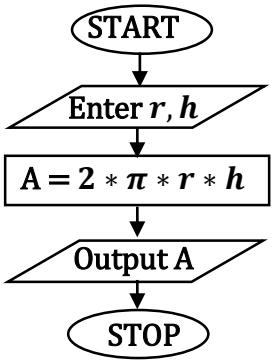
Stop.

2021.P1.Nov.

In the answer space below is an incomplete flowchart for calculating the curved surface area (A) of a cylinder with radius (r) and height (h). complete the flowchart.



Solution



2022.P1.Nov

In the answer space below is an incomplete program written in pseudocode for calculating the curved surface area (A) of a cone with base radius (r) and slant height (s). Complete the program. ($A = \pi rs$)

```
Start
    Enter ……..
    A = ……..
    Output A
Stop.
```

Solution

```
Start
    Enter r, s
    A = π * r * s
    Output A
Stop.
```

2023.P1.Nov

In the answer space below is an incomplete simple program in pseudocode for calculating and outputting the volume, V, of a cone given the base radius r and height h. Complete the program by filling in the blank spaces with appropriate statements.

```
Begin
    Enter ……..
    V = ……..
    Output V
End.
```

Solution

```
Begin
    Enter r, h
    V = 1/3 * π * r² * h
    Output V
End.
```

2024.P1.GCE

In the answer space below is an incomplete program written in pseudocode for calculating the nth term (T_n) of an arithmetic progression, given the first term (a), the common difference (d) and the number of terms (n). Complete the program.

Start

Enter .....

T_n = ...

Output T_n

Stop.

Solution

Start

Enter a, d, n

$T_n = a + (n - 1) * d$

Output T_n

Stop.

2018.P2.GCE

Study the pseudo code below.

Start

Enter a, r, n

$R = 1 - r$

If $R = 0$ THEN

Print "the value of r is not valid"

Else $S_n = \frac{a(1-r^n)}{R}$

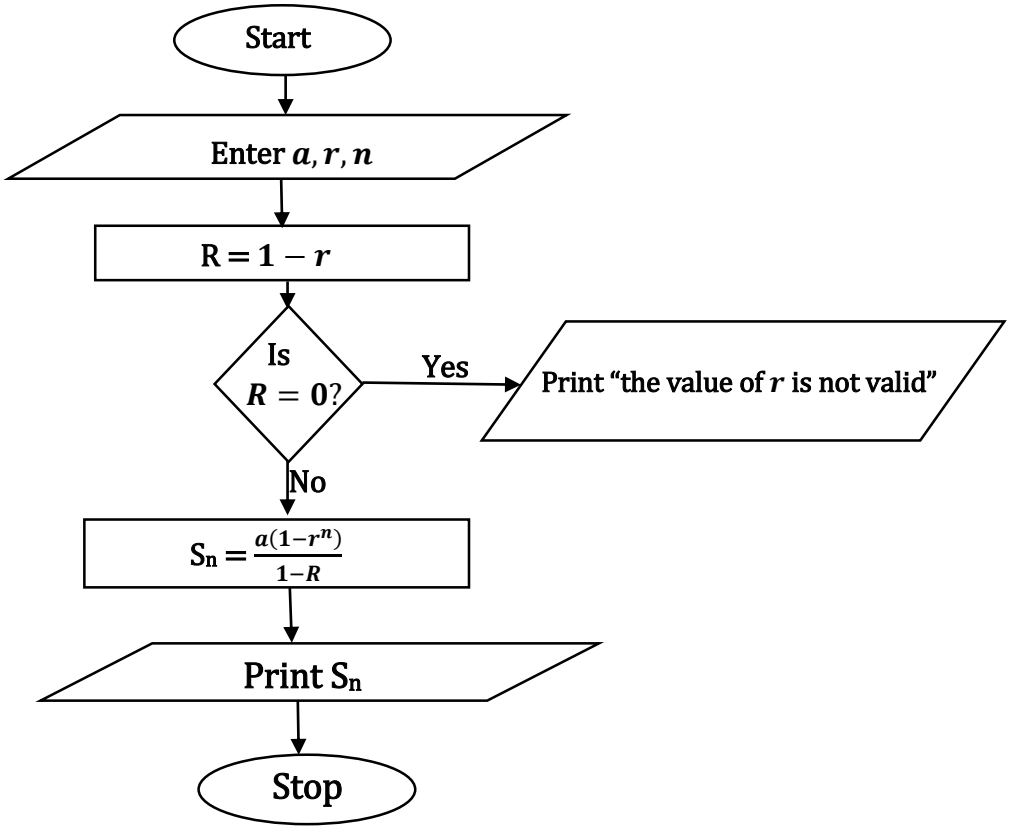
End if

Print S_n

Stop

Construct a flow chart corresponding to the pseudo code above. [5]

Solution



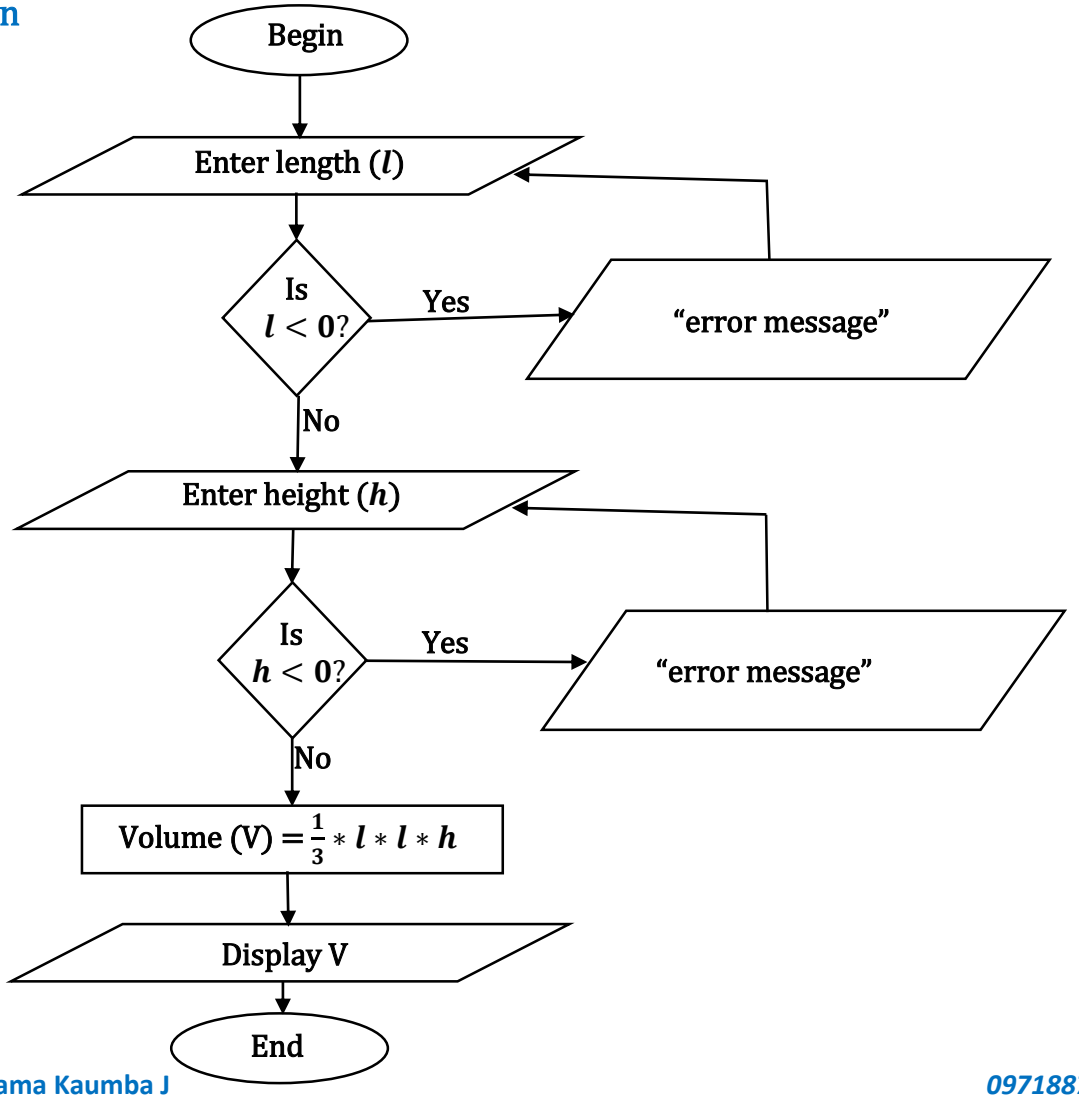
2019.P2.Nov

The program below is given in the form of a pseudo code.

Begin
  Enter length
    If length < 0
      Then display "error message" and re-enter positive length
    Else enter height
    If height < 0
      Then display "error message" and re-enter positive height
   Else volume $= \frac{1}{3} * l * l * h$
      End if
   Display volume
End

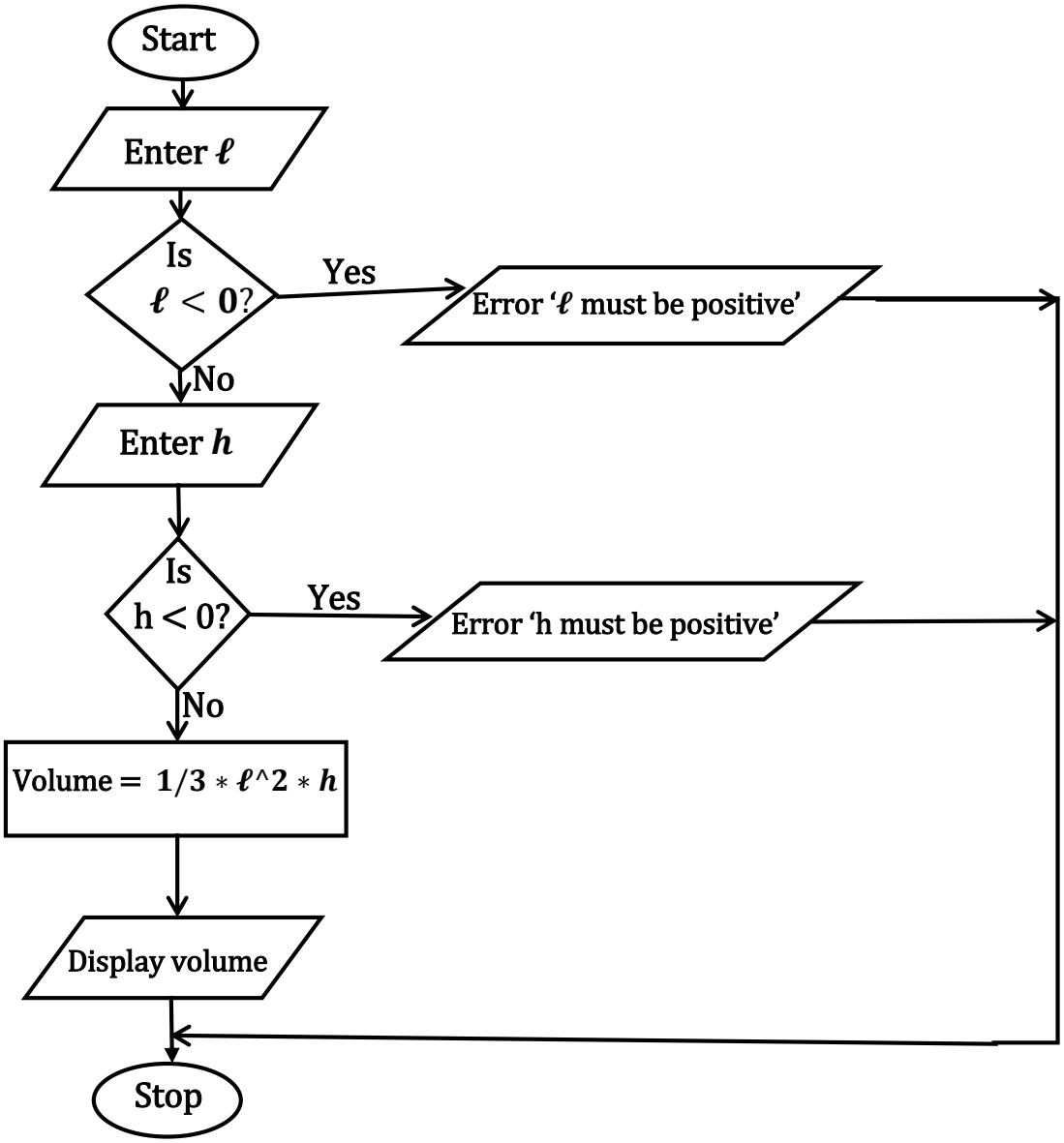
Draw the corresponding flowchart for the information given above. [5]

Solution



2020.P2.Nov

Study the flow chart below.



Write a pseudo code corresponding to the flow chart programme above. [5]

Solution

Start

Enter ℓ

If $\ell < 0$

Then display error 'ℓ must be positive'

Else enter h

If $h < 0$

Then display error 'h must be positive'

Else Volume $= 1/3 * \ell$^2 $* h$

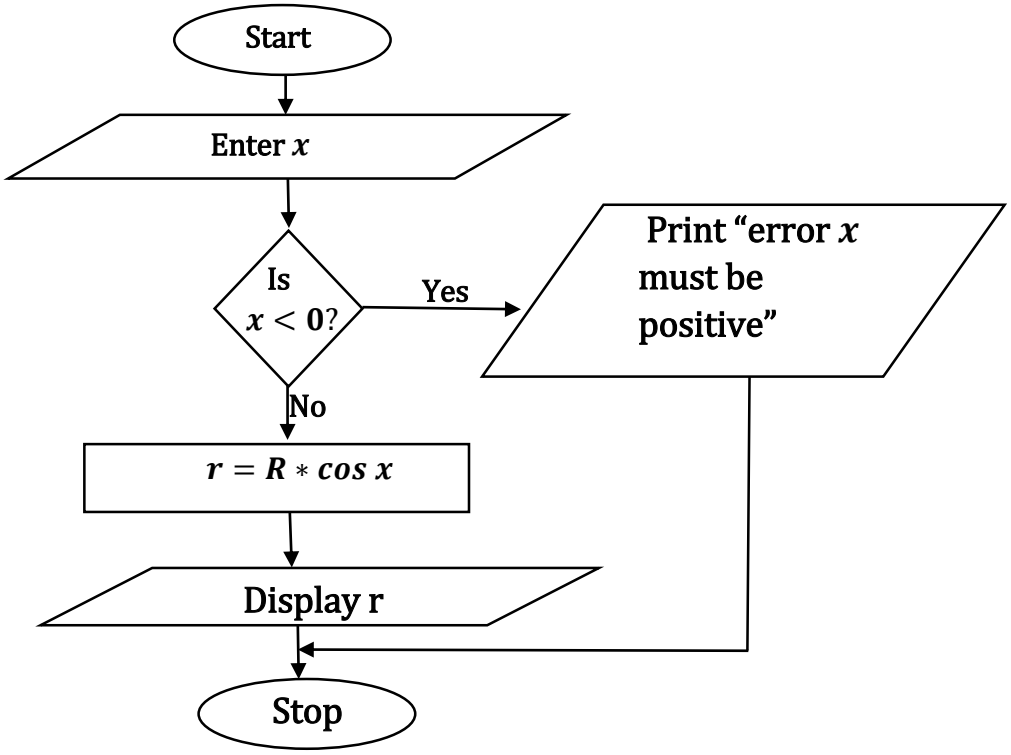
End if

Display volume

Stop.

2021.P1.Nov

Study the flowchart below.



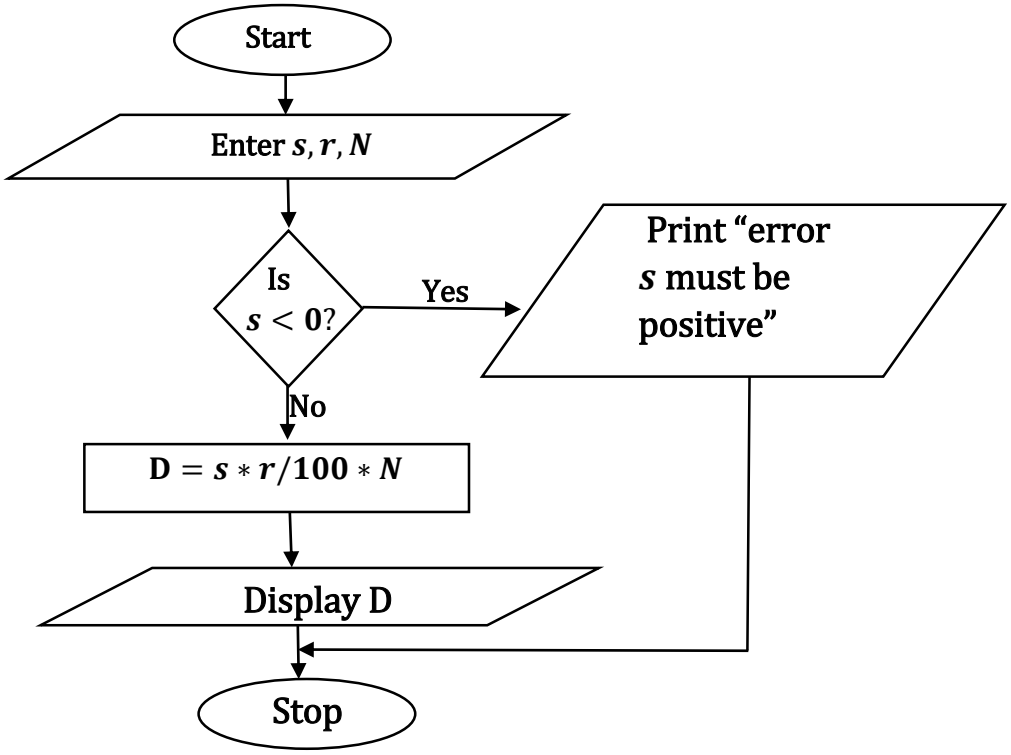
Write a pseudocode corresponding to the flowchart programme. [5]

Solution

```
Start
    Enter x
      If x < 0
      Then Print "error x must be positive"
     Else r = R * cos x
   End if
  Display, r
```

2022.P2.Nov

Study the flowchart below.



Write a pseudocode corresponding to the flowchart above. [5]

Solution

```
Start
    Enter s, r, N
      If s < 0
      Then Print "Error s must be positive"
     Else D = s * r/100 * N
    End if
  Display D
Stop.
```

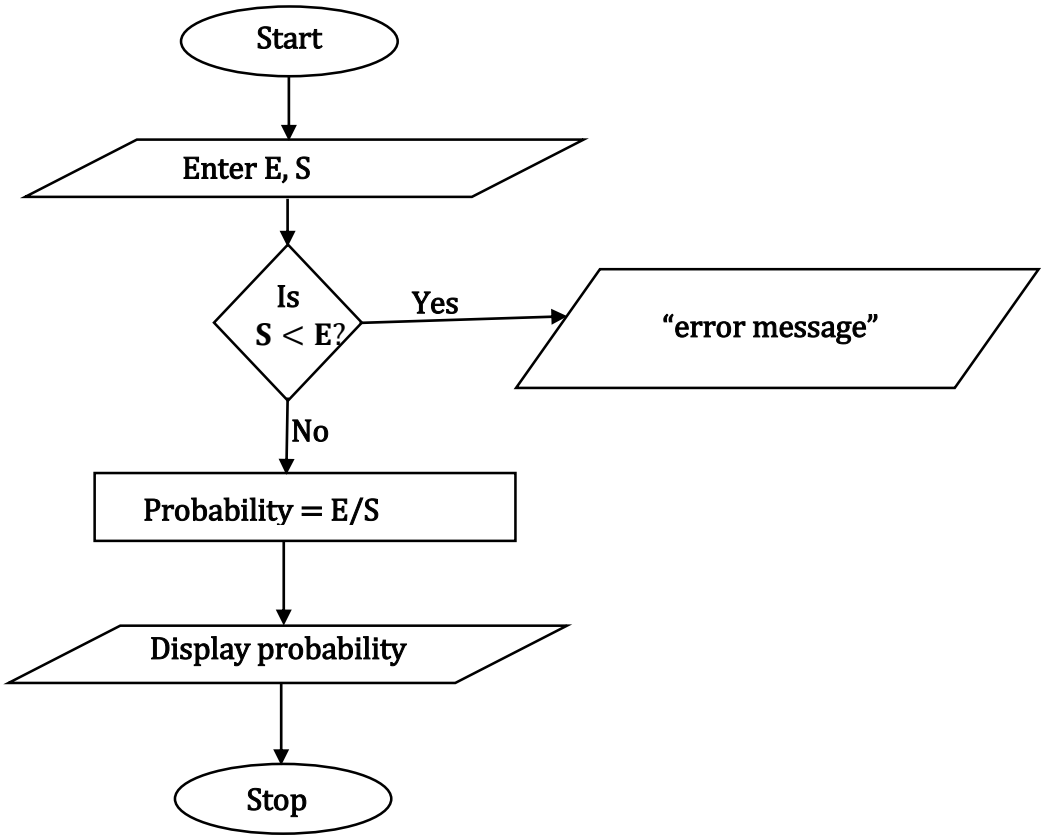
2023.P2.Nov

Study the following pseudocode.

```
Start
    Enter E, S
        IF S < E THEN
         Display 'error message'
         ELSE probability = E/S
         END IF
     Display probability
Stop
```

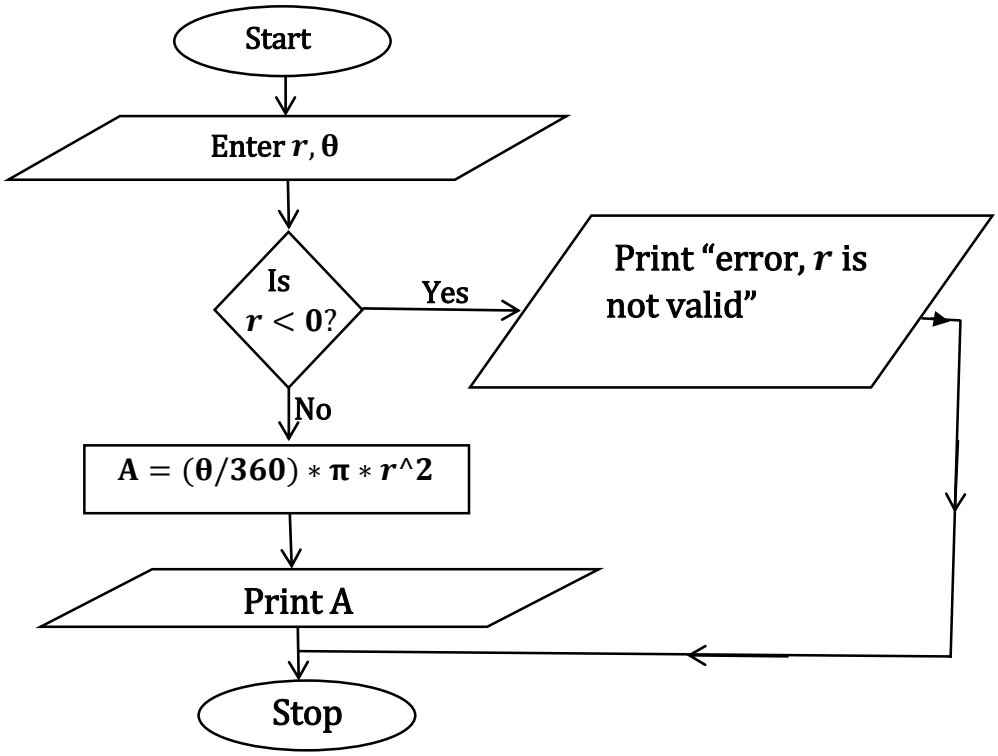
Draw a corresponding flowchart for the pseudocode above. [5]

Solution



2024.P2.GCE

The flowchart below shows steps that are carried out in calculating and displaying the area (A) of a sector given its radius (r) and angle of the sector (θ).



Write a pseudocode corresponding to the flowchart program above. [5]

Solution

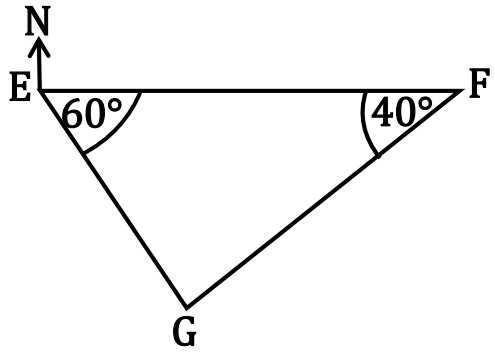
```
Start
    Enter r, θ
      If r < 0
      Then Print "error, r is not valid" and stop.
     Else A = (θ/360) * π * r ∧ 2
    End if
  Print A
Stop.
```

BEARINGS

- A bearing is an angle measured in a clockwise direction from a North pole to a given fixed point.
- It is always stated in three (3) figures.

2018.P1.Oct

In the diagram below, F is due east of E, angle GEF = 60° and angle EFG = 40°.



Calculate the bearing of

(a) F from G,

(b) E from G.

Solution

(a) $\text{F from G} = 180 - (90 + 40)$

$= 180 - 130$

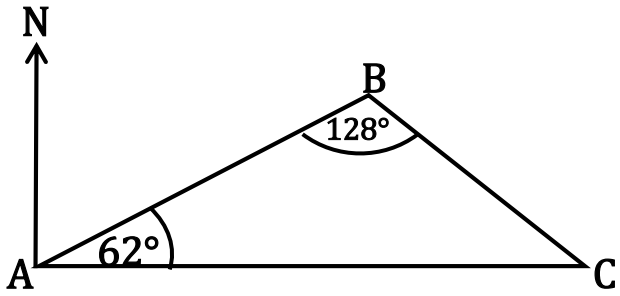
$= 050°$

(b) $\text{E from G} = 360 - 30$

$= 330°$

2019.P1.Nov

In the diagram below, A, B and C are three points on level ground. The bearing of B from A is 062° and angle ABC =128°. C is due east of A.



Find the bearing of

(a) C from B,

(b) A from C.

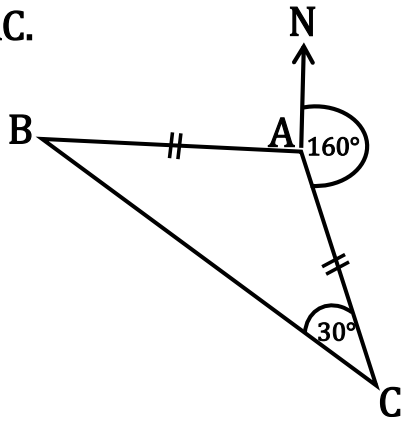
Solution

(a) C from B $= 360 - (118 + 128)$
$= 360 - 246$
$= 114°$

(b) A from C $= 360 - 90$
$= 270°$

2020.P1.Nov

In the diagram below, the bearing of C from A is 160°. Angle ACB = 30° and AB = AC.



Find

(a) angle BAC,

(b) the bearing of A from C.

Solution

(a) $B\hat{A}C = 180 - (30 + 30)$

$= 180 - 60$

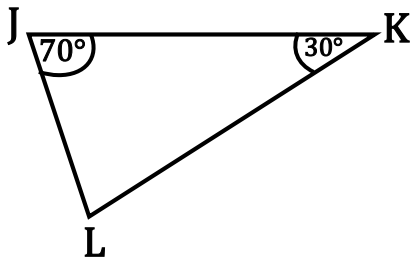
$= 120°$

(b) A from C $= 360 - 20$

$= 340°$

2021.P1.Nov

The diagram below shows three points J, K and L on level ground. K is due east of J, angle LJK = 70° and angle JKL = 30°.



Calculate the bearing of

(a) K from L,

(b) J from L.

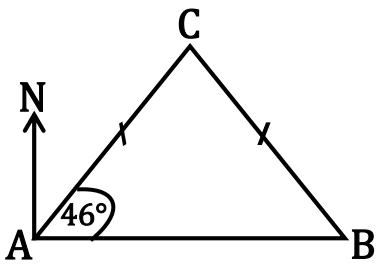
Solution

(a) K from L $= 180 - (90 + 30)$
$= 180 - 120$
$= 060°$

(b) J from L $= 360 - [180 - (90 + 70)]$
$= 360 - (180 - 160)$
$= 360 - 20$
$= 340°$

2022.P1.Nov

In the following diagram, angle BAC is 46° and AC = BC. B is due east of A.



Calculate the bearing of

(a) A from C,

(b) C from B.

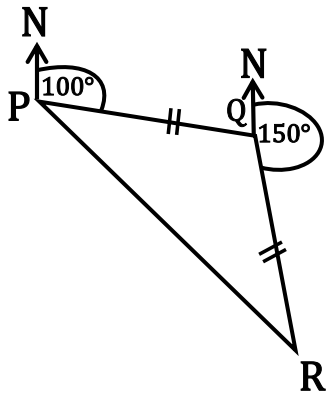
Solution

(a) $\text{A from C} = 360 - [180 - (90 - 46)]$
$= 360 - (180 - 44)$
$= 360 - 136$
$= 224°$

(b) $\text{C from B} = 360 - 44$
$= 316°$

2023.P1.Nov

The diagram below shows three points P, Q and R on level ground. The bearing of Q fro P is 100°, the bearing of R from Q is 150° and PQ = QR.



Find

(a) angle QPR,

(b) the bearing of P from R.

Solution

(a) $\widehat{QPR} = \frac{180-[360-(80+150)]}{2}$

$= \frac{180-(360-230)}{2}$

$= \frac{180-130}{2}$

$= \frac{50}{2}$

$= 22°$

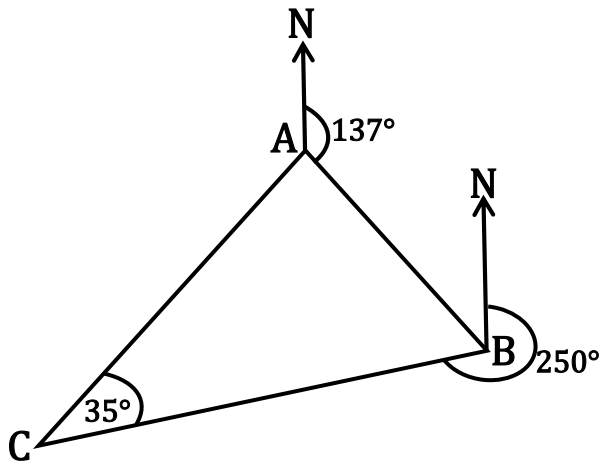
(b) P from R $= 360 - (30 + 22)$

$= 360 - 52$

$= 308°$

2024.P1.GCE

In the diagram below, A, B and C are three points on level ground. The bearing of B from A is 137°, the bearing of C from B is 250° and angle ACB = 35°.



Find the bearing of

(a) A from B,

(b) A from C.

Solution

(a) $\text{A from B} = 360 - (180 - 137)$
$= 360 - 43$
$= 317°$

(b) $A\widehat{B}C = 360 - (250 + 43)$
$= 360 - 293$
$= 67°$
$B\widehat{A}C = 180 - (35 + 67)$
$= 180 - 102$
$= 78°$
$\therefore \text{A from C} = 180 - [360 - (137 + 78)]$
$= 180 - (360 - 215)$
$= 180 - 145$
$= 035°$

TRAVEL GRAPHS

- Travel graphs are graphical representations of distance-time relationships. They are used to analyze and visualize the motion of objects.

Types of Travel Graphs

1. Distance -Time Graphs: Shows the distance traveled over time.
2. Speed-Time Graphs: Shows the speed of an object over time.
3. Acceleration-Time Graphs: Show the acceleration of an object over time.

Interpreting Travel Graphs

1. Distance-Time Graphs: A straight line indicates constant speed, while a curved line indicates changing speed.
2. Speed-Time Graphs: A straight line indicates constant acceleration, while a curved line indicates changing acceleration.
3. Acceleration-Time Graphs: A straight line indicates constant acceleration, while a curved line indicates changing acceleration.

Velocity, Acceleration and Retardation

- Velocity, acceleration, and retardation are fundamental concepts in mathematics and physics that describe the motion of objects.

Velocity

- Velocity is the rate of change of an object's position with respect to time.
- It is a vector quantity, having both magnitude (amount of movement) and direction.

Types of Velocity

1. Uniform Velocity: Velocity that remains constant over time.
2. Non-Uniform Velocity: Velocity that changes over time.
3. Instantaneous Velocity: Velocity at a specific instant in time.

Equation of Velocity

- $\mathbf{v} = \frac{\Delta s}{\Delta t}$

Acceleration

- Acceleration is the rate of change of velocity.
- It is a measure of how quickly an object's velocity changes.

Types of Acceleration

1. Uniform Acceleration: Acceleration that remains constant over time.
2. Non-Uniform Acceleration: Acceleration that changes over time.
3. Instantaneous Acceleration: Acceleration at a specific instant in time.

Equation of Acceleration

- $a = \frac{\Delta v}{\Delta t}$

Retardation

- Retardation is the opposite of acceleration.
- It is the rate of decrease of velocity.

Types of Retardation

1. Uniform Retardation: Retardation that remains constant over time.
2. Non-Uniform Retardation: Retardation that changes over time.
3. Instantaneous Retardation: Retardation at a specific instant in time.

Equation of Retardation

- $r = -a$

Velocity-Time Graphs:

- A velocity-time graph is a graphical representation of an object's velocity over time.
- It is a powerful tool for analyzing and understanding motion.

Interpreting a Velocity-Time Graph

1. Velocity: The y-axis represents velocity, with positive values indicating motion in one direction and negative values indicating motion in the opposite direction.
2. Time: The x-axis represents time, with the origin (0, 0) representing the starting point.
3. Gradient: The gradient of the graph represents acceleration, with a positive gradient indicating acceleration and a negative gradient indicating retardation.
4. Area: The area under the graph represents displacement (distance), with the sign of the area indicating the direction of displacement.

Types of Velocity-Time Graphs

1. Uniform Velocity: A horizontal line indicates uniform velocity.
2. Uniform Acceleration: A straight line with a positive gradient indicates uniform acceleration.
3. Uniform Retardation: A straight line with a negative gradient indicates uniform retardation.
4. Non-Uniform Motion: A curved line indicates non-uniform motion.

Constant Velocity

- Constant velocity occurs when an object moves in a straight line at a constant speed.
- This means that the object's velocity does not change over time.

Characteristics of Constant Velocity

1. Straight-Line Motion: The object moves in a straight line.
2. Constant Speed: The object's speed does not change over time.
3. Zero Acceleration: The object's acceleration is zero, since its velocity is not changing.

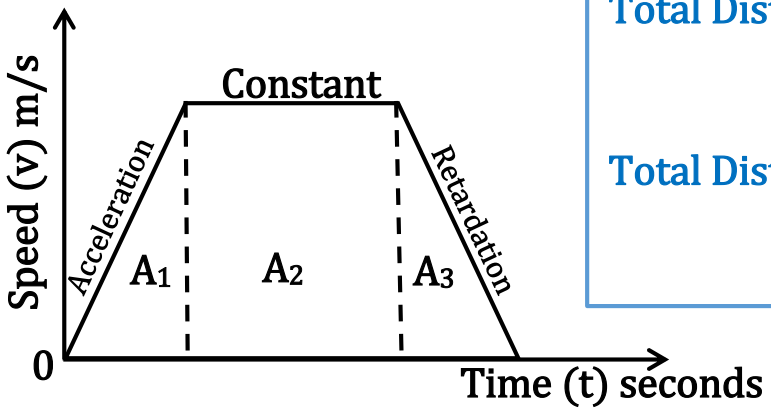
Graphical Representation

- A constant velocity is represented by a horizontal line on a velocity-time graph.

Equations of Motion

- For an object moving with constant velocity:

1. $v = u$: The final velocity (v) is equal to the initial velocity (u).
2. $s = vt$: The displacement (s) is equal to the product of the velocity (v) and time (t).
3. $a = 0$: The acceleration (a) is zero.



Total Distance $= A_1 + A_2 + A_3$

Or

Total Distance $= \frac{1}{2}(a + b)h$

Distance under a Velocity-Time Graph

- The area under a velocity-time graph represents the distance traveled by an object.

Understanding the Concept

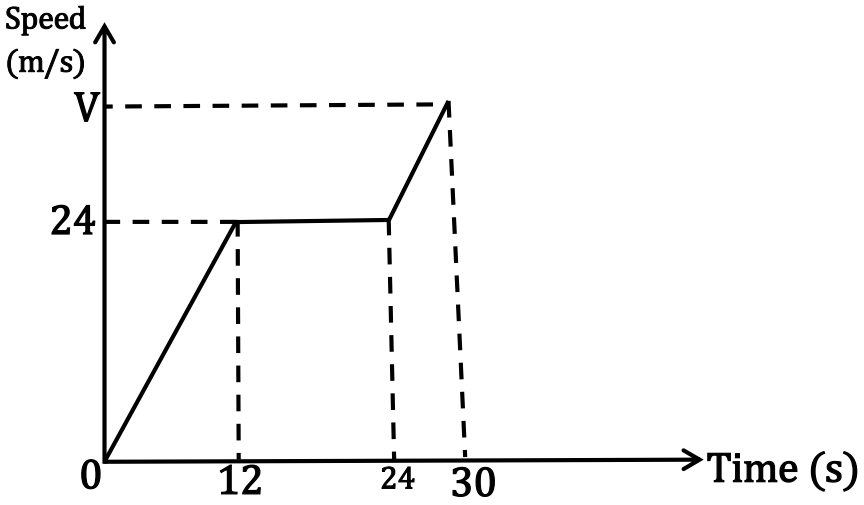
1. Area under the graph: The area under the velocity-time graph represents the distance traveled.
2. Positive area: A positive area under the graph represents a distance traveled in the positive direction.
3. Negative area: A negative area under the graph represents a distance traveled in the negative direction.

Calculating Distance

1. Rectangle method: Divide the area under the graph into rectangles and calculate the area of each rectangle.
2. Trapezium method: Divide the area under the graph into trapeziums and calculate the area of each trapezium.
3. Integration method: Use integration to calculate the area under the graph.

2018.P1.Oct

The diagram below is a speed-time graph of a particle which accelerates uniformly from rest for 12 second until it reaches a speed of 24m/s. it moves at a constant speed for a further 12 seconds before it accelerates uniformly for another 6 seconds to a speed of V m/s.



(a) Calculate the acceleration for the first 12 seconds.

(b) Find the distance which the particle covered in the first 24 seconds.

(c) Given that the total distance covered was 600m, calculate the value of V.

Solution

(b) $a = \frac{v-u}{t}$

$a = \frac{24-0}{12}$

$a = \frac{24}{12}$

$a = 2m/s^2$

(c) $D = \frac{1}{2}(a+b)h$

$= \frac{1}{2}(12+24)24$

$= 12(36)$

$= 432m$

(a) $432 + \frac{1}{2}(24+V)6 = 600$

$432 + 3(24+V) = 600$

$432 + 72 + 3V = 600$

$504 + 3V = 600$

$3V = 600 - 504$

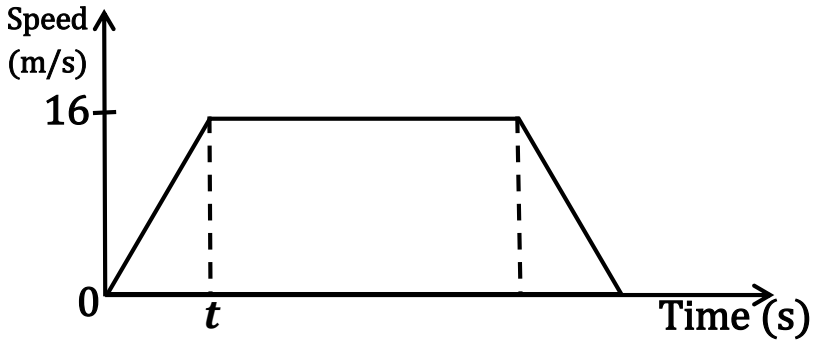
$3V = 96$

$V = \frac{96}{3}$

$V = 32m/s$

2019.P1.Nov

The diagram below is the speed time graph of a car. The car starts from rest and accelerates uniformly at $2m/s^2$ for t seconds until it reaches a speed of 16m/s. It then travels at 16m/s for 40 seconds, after which it comes to rest in further 10 seconds.



Find the

(a) value of t,

(b) distance travelled in the last 50 seconds,

(c) speed of the car when $t = 53$ secnds.

Solution

(b) $a = \frac{v-u}{t}$

$2 = \frac{16-0}{t}$

$2t = 16$

$t = \frac{16}{2}$

$t = 8\ seconds$

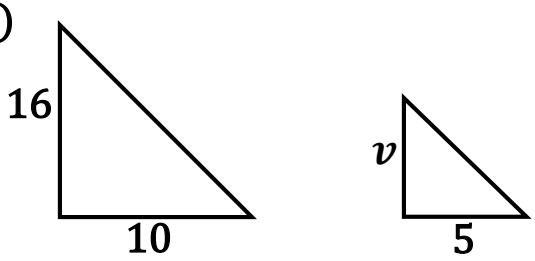
(c) $D = \frac{1}{2}(a+b)h$

$= \frac{1}{2}(40+50)16$

$= 8(90)$

$= 720m$

(a)



$16 : v = 10 : 5$

$16 : v = 2 : 1$

$\frac{16}{v} = \frac{2}{1}$

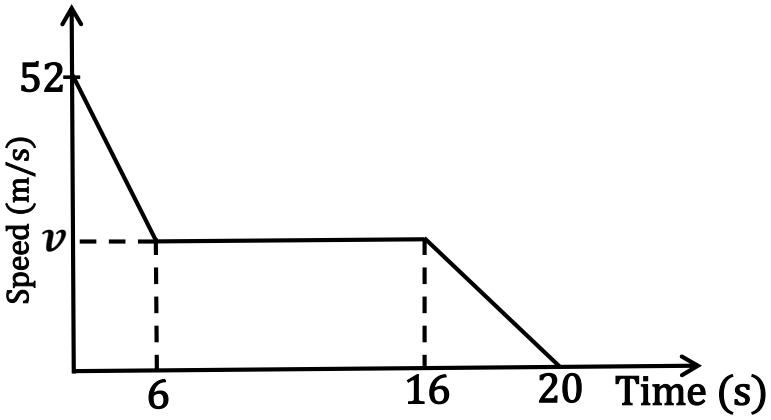
$2v = 16$

$v = \frac{16}{2}$

$v = 8m/s$

2020.P1.Nov

The diagram below is a graph of a moving object which decelerates uniformly at 8m/s² from a speed of 52m/s to vm/s in 6 seconds. It then moves at a constant speed for 10 seconds and finally decelerates uniformly to come to rest in 4 seconds.



Calculate the

(a) speed v,

(b) total distance travelled for the whole journey,

(c) average speed for the whole journey.

Solution

(a) $a = \frac{v-u}{t}$

$8 = \frac{52-V}{6}$

$48 = 52 - v$

$v = 52 - 48$

$v = 4/ms$

(b) $D = \frac{1}{2}(a+b)h + \frac{1}{2}(a+b)h$

$= \frac{1}{2}(4+52)6 + \frac{1}{2}(10+14)4$

$= 168 + 48$

$= 216m$

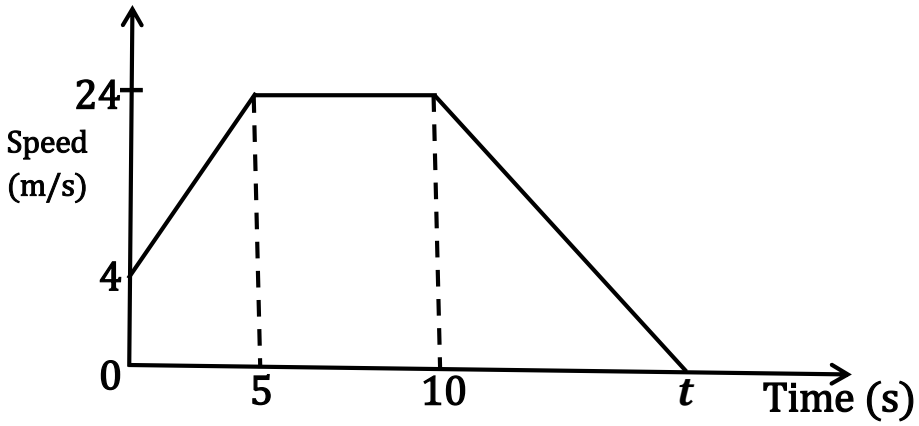
(c) $S = \frac{D}{T}$

$= \frac{216}{20}$

$= 10\frac{4}{5}m/s$

2021.P1.Nov

The diagram below shows a speed-time graph of a moving object.



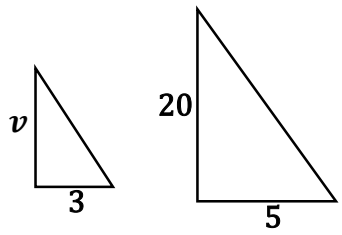
(a) Find the acceleration of the object when $t = 3$ seconds.

(b) Given that the object decelerates at $2m/s^2$, find the value of t.

(c) Calculate the average speed of the object in the first 10 seconds of the journey.

Solution

(c)



$3 : 5 = v : 20$

$\frac{3}{5} = \frac{v}{20}$

$5v = 20 \times 3$

$v = \frac{60}{5}$

$v = 12m/s$

$\therefore a = \frac{v}{t}$

$a = \frac{12}{3}$

$a = 4m/s^2$

(a) $2 = \frac{24}{t-10}$

$2(t - 10) = 24$

$2t - 20 = 24$

$2t = 24 + 20$

$2t = 44$

$t = \frac{44}{2}$

$t = 22\ seconds$

(b) $D = \frac{1}{2}(a + b)h + lb$

$D = \frac{1}{2}(4 + 24)5 + 24 \times 5$

$D = \frac{1}{2}(28)5 + 120$

$D = 70 + 120$

$D = 190m$

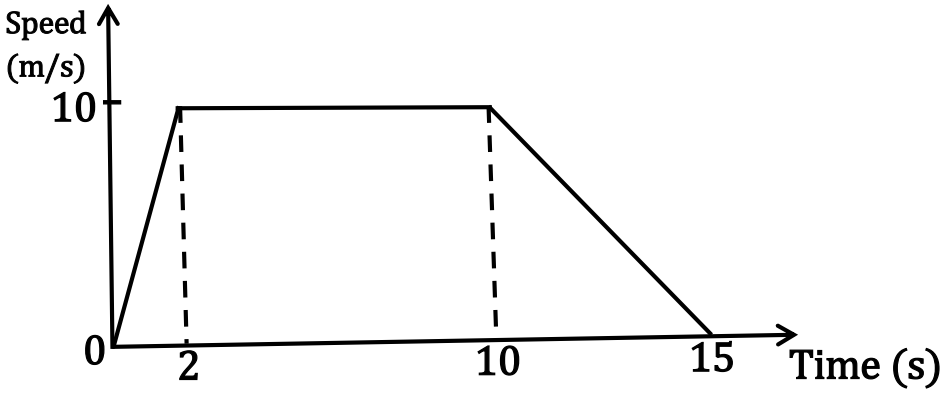
$S = \frac{D}{T}$

$S = \frac{190}{10}$

$A.S = 19m/s$

2022.P1.Nov

The diagram below is the speed time graph of an object. The object starts from rest and accelerates uniformly for 2 seconds until it reaches a speed of 10m/s. It then travels at this speed for 8 seconds and finally decelerates to rest after 5 seconds.



Find the

(a) retardation of the object in the last 5 seconds,

(b) distance travelled in the first 10 seconds,

(c) average speed of the object for the whole journey.

Solution

(a) $r = \frac{u-v}{t}$

$r = \frac{0-10}{15-10}$

$r = \frac{-10}{5}$

$r = -2m/s^2$

(b) $\mathrm{D} = \frac{1}{2}(a+b)h$

$\mathrm{D} = \frac{1}{2}(8+10)10$

$\mathrm{D} = 5(18)$

$\mathrm{D} = 90m$

(c) $\mathrm{D} = 90 + \frac{1}{2} \times 5 \times 10$

$\mathrm{D} = 90 + 25$

$\mathrm{D} = 115m$

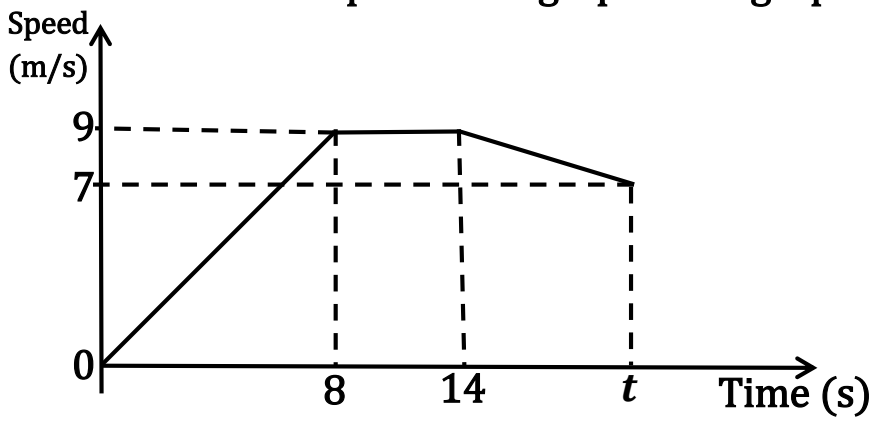
$\therefore A.S = \frac{D}{T}$

$= \frac{115}{15}$

$= 7\frac{2}{3}m/s$

2023.P1.Nov

The diagram below shows the speed-time graph during a period of t seconds.



Find the

(a) acceleration of the car during the first 8 seconds,

(b) distance covered in the first 14 seconds,

(c) value of t if the average speed of the car is 7m/s.

Solution

(b) $a = \frac{v-u}{t}$

$a = \frac{9-0}{8}$

$a = \frac{9}{8}$

$a = 1\frac{1}{8} m/s^2$

(c) $D = \frac{1}{2}(a+b)h$

$D = \frac{1}{2}(6+14)9$

$D = 10 \times 9$

$D = 90m$

(a) $T.D = 90 + \frac{1}{2}(7+9)(t-14)$

$= 90 + 8(t-14)$

$= 90 + 8t - 112$

$= 8t + 90 - 112$

$= 8t - 22$

$A.S = \frac{T.D}{T}$

$7 = \frac{8t-22}{t}$

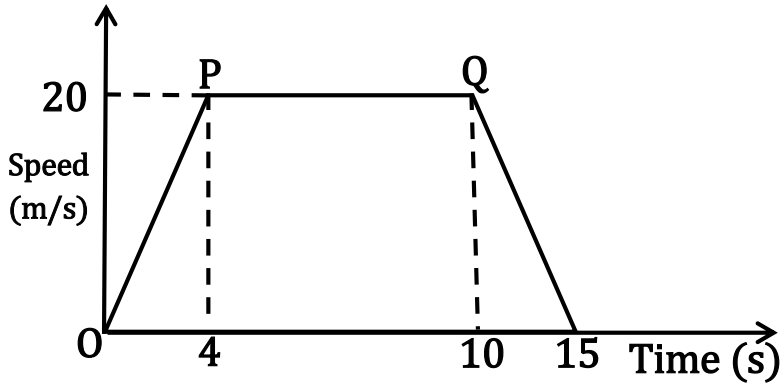
$8t - 22 = 7t$

$8t - 7t = 22$

$t = 22\ seconds$

2024.P1.GCE

The diagram below shows the speed-time graph of an object. The object accelerates uniformly from O to P until it reaches a speed of 20m/s in 4 seconds and then moves steadily from P to Q for 6 seconds and finally comes to rest.



Find the

(a) acceleration in the first 4 seconds,

(b) total distance covered by the object,

(c) average speed in the last 4 seconds.

Solution

(b) $a = \frac{v-u}{t}$

$= \frac{20-0}{4}$

$= \frac{20}{4}$

$= 5m/s^2$

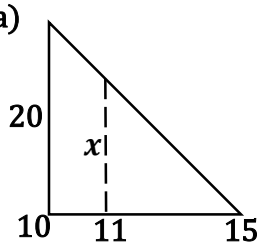
(c) $D = \frac{1}{2}(a+b)h$

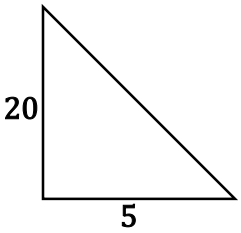
$= \frac{1}{2}(6+15)20$

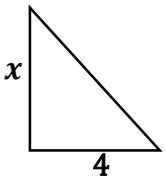
$= 10(21)$

$= 210m$

(a)







$20 : x = 5 : 4$

$\frac{20}{x} = \frac{5}{4}$

$5x = 20 \times 4$

$5x = 80$

$x = \frac{80}{5}$

$x = 16$

$D = \frac{1}{2}bh$

$= \frac{1}{2} \times 4 \times 16$

$= 32m$

$\therefore A.S = \frac{D}{T}$

$= \frac{32}{4}$

$= 8m/s$